19

Temperature

CHAPTER OUTLINE

19.1 Temperature and the Zeroth Law of Thermodynamics

19.2 Thermometers and the Celsius Temperature Scale

19.3 The Constant-Volume Gas Thermometer and the Absolute Temperature Scale

19.4 Thermal Expansion of Solids and Liquids

19.5 Macroscopic Description of an Ideal Gas

* An asterisk indicates a question or problem new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ19.1 Answer (b). The markings are now farther apart than intended, so measurements made with the heated steel tape will be too short—but only by a factor of 5×10^{-5} of the measured length.

OQ19.2 Answer (d). Remember that one must use absolute temperatures and pressures in the ideal gas law. Thus, the original temperature is $T_K = T_C + 273.15 = 25 + 273.15 = 298$ K, and with the mass of the gas constant, the ideal gas law gives

$$T_2 = \left(\frac{P_2}{P_1}\right)\left(\frac{V_2}{V_1}\right)T_1 = \left(\frac{1.07 \times 10^6 \text{ Pa}}{5.00 \times 10^6 \text{ Pa}}\right)(3.00)(298 \text{ K}) = 191 \text{ K}$$

OQ19.3 Answer (d). From the ideal gas law, with the mass of the gas constant, $P_2V_2/T_2 = P_1V_2/T_1$. Thus,

$$P_2 = \left(\frac{V_1}{V_2}\right)\left(\frac{T_2}{T_1}\right)P_1 = \left(\frac{1}{2}\right)(4)P_1 = 2P_1$$

OQ19.4 Answer (a). As the temperature increases, the brass expands. This would effectively increase the distance d from the pivot point to the

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center of mass of the pendulum, and also increase the moment of inertia of the pendulum. Since the moment of inertia is proportional to d^2, and the period of a physical pendulum is $T = 2\pi\sqrt{\dfrac{I}{mgd}}$, the period would increase, and the clock would run slow.

OQ19.5 Answer (c). $T_C = \dfrac{5}{9}(T_F - 32) = \dfrac{5}{9}(162 - 32) = 72.2°\text{C},$ then,

$$T_K = T_C + 273.15 = 72.2 + 273.15 = 345 \text{ K}$$

OQ19.6 Answer (c). From the ideal gas law, with the mass of the gas constant, $P_2V_2/T_2 = P_1V_2/T_1$. Thus,

$$V_2 = \left(\frac{P_1}{P_2}\right)\left(\frac{T_2}{T_1}\right)V_1 = (4)(1)\left(0.50 \text{ m}^3\right) = 2.0 \text{ m}^3$$

OQ19.7 Answer (d). If glass were to expand more than the liquid, the liquid level would fall relative to the tube wall as the thermometer is warmed. If the liquid and the tube material were to expand by equal amounts, the thermometer could not be used because the liquid level would not change with temperature.

OQ19.8 The ranking is (a) = (b) = (d) > (e) > (c). We think about nRT/V in each case. Since R is constant, we need only think about nT/V, and units of mmol·K/cm^3 are as convenient as any: (a) 2·3/1 = 6, (b) 6, (c) 4, (d) 6, (e) 5.

OQ19.9 Answer (d). Cylinder A must be at lower pressure. If the gas is thin, $PV = nRT$ applies to both with the same value of nRT for both. Then A will be at one-third the absolute pressure of B.

OQ19.10 (i) Answer (a). Call the process isobaric cooling or isobaric contraction. The rubber wall is easy to stretch. The air inside is nearly at atmospheric pressure originally and stays at atmospheric pressure as the wall moves in, just maintaining equality of pressure outside and inside. The air is nearly an ideal gas to start with, and stays fairly ideal—fairly far from liquefaction—even at 100 K. The water vapor liquefies and then freezes, and the carbon dioxide turns to dry ice, but these are minor constituents of the air. Thus, as the absolute temperature drops to 1/3 of its original value and the volume will drop to 1/3 of what it was.

(ii) Answer (c). As noted above, the pressure stays nearly constant at 1 atm.

OQ19.11 Answer (c). For a quick approximation, multiply 93 m and 17 and

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1/(1 000 000 °C) and say 5°C for the temperature increase. To simplify, multiply 100 and 100 and 1/1 000 000 for an answer in meters: it is on the order of 1 cm.

OQ19.12 Answer (b). Around atmospheric pressure, 0°C is the only temperature at which liquid water and solid water can both exist.

OQ19.13 Answer (b). When a solid, containing a cavity, is heated, the cavity expands in the same way as it would if filled with the material making up the rest of the object.

OQ19.14 Answer (e).

$$T_F = \frac{9}{5}T_C + 32 = \frac{9}{5}(-25°) + 32° = -13° \text{ F}$$

ANSWERS TO CONCEPTUAL QUESTIONS

CQ19.1 The coefficient of linear expansion must be greater for mercury than for glass, otherwise the interior of a glass thermomter would expand more and the mercury level would drop. See OQ19.7.

CQ19.2 (a) The copper's temperature drops and the water temperature rises until both temperatures are the same. (b) The water and copper are in thermal equilibrium when their temperatures are the same.

CQ19.3 (a) $PV = nRT$ predicts V going to zero as T goes to zero.

(b) The ideal-gas model does not apply when the material gets close to liquefaction and then turns into a liquid or solid. The molecules start to interact all the time, not just in brief collisions. The molecules start to take up a significant portion of the volume of the container.

CQ19.4 Air pressure decreases with altitude while the pressure inside the bags stays the same; thus, that inside pressure is greater than the outside pressure.

CQ19.5 (a) No. The thermometer will only measure the temperature of whatever is in contact with the thermometer. The thermometer would need to be brought to the surface in order to measure its temperature, since there is no atmosphere on the Moon to maintain a relatively consistent ambient temperature above the surface. (b) It would read the temperature of the glove, since it is in contact with the glove.

CQ19.6 The coefficient of expansion of metal is larger than that of glass. When hot water is run over the jar, both the glass and the lid expand, but at different rates. Since *all* dimensions expand, the inner

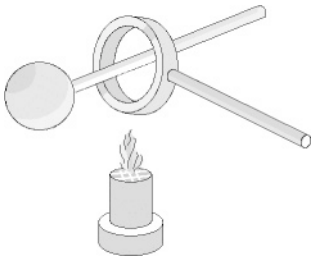
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diameter of the lid expands more than the top of the jar, and the lid will be easier to remove.

CQ19.7 (a) As the water rises in temperature, it expands or rises in pressure or both. The excess volume would spill out of the cooling system, or else the pressure would rise very high indeed.

(b) Modern cooling systems have an overflow reservoir to accept the excess volume when the coolant heats up and expands.

CQ19.8 (a) The sphere expands when heated, so that it no longer fits through the ring. With the sphere still hot, you can separate the sphere and ring by heating the ring. This more surprising result occurs because the thermal expansion of the ring is not like the inflation of a blood-pressure cuff. Rather, it is like a photographic enlargement; every linear dimension, including the hole diameter, increases by the same factor. The reason for this is that the atoms everywhere, including those around the inner circumference, push away from each other. The only way that the atoms can accommodate the greater distances is for the circumference—and corresponding diameter—to grow. This property was once used to fit metal rims to wooden wagon wheels. If the ring is heated and the sphere left at room temperature, the sphere would pass through the ring with more space to spare.



ANS. FIG. CQ19.8

(b) Heating the ring increases its diameter, the sphere can pass through it easily. The hole in the ring expands as if it were filled with the material of the ring.

CQ19.9 Two objects in thermal equilibrium need not be in contact. Consider the two objects that are in thermal equilibrium in Figure 16.1(c). The act of separating them by a small distance does not affect how the molecules are moving inside either object, so they will still be in thermal equilibrium.

CQ19.10 (a) One mole of H_2 has a mass of 2.016 0 g.

(b) One mole of He has a mass of 4.002 6 g.

(c) One mole of CO has a mass of 28.010 g.

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SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 19.2 Thermometers and the Celsius Temperature Scale

Section 19.3 The Constant-Volume Gas Thermometer and the Absolute Temperature Scale

P19.1 (a) By Equation 19.2,

$$T_F = \frac{9}{5}T_C + 32 = \frac{9}{5}(41.5°\text{C}) + 32 = (74.7 + 32)°\text{F} = \boxed{107°\text{F}}$$

(b) Yes. The normal body temperature is 98.6°F, so the patient has a high fever and needs immediate attention.

P19.2 (a) Consider the freezing and boiling points of water in each scale: 0°C and 100°C; 32°F and 212°F. We see that there are 100 Celsius units for every 180 Fahrenheit units:

$$\frac{\Delta T_C}{\Delta T_F} = \frac{100°\text{C}}{180°\text{F}} \quad \rightarrow \quad \Delta T_C = \frac{5}{9}(\Delta T_F) = \frac{5}{9}(57.0)°\text{C} = \boxed{31.7°\text{C}}$$

(b) The Kelvin unit is the same size as the Celsius unit:

$$T = T_C + 273.15 \quad \rightarrow \quad \Delta T = \Delta T_C$$

$$\Delta T = \Delta T_C\left(\frac{1\text{ K}}{1\text{ °C}}\right) = 31.7\text{ K}\left(\frac{1\text{ K}}{1\text{ °C}}\right) = \boxed{31.7\text{ K}}$$

P19.3 (a) By Equation 19.2,

$$T_F = \frac{9}{5}T_C + 32 = \frac{9}{5}(-78.5) + 32 = \boxed{-109°\text{F}}$$

And, from Equation 19.1,

$$T = T_C + 273.15 = (-78.5 + 273.15)\text{ K} = \boxed{195\text{ K}}$$

(b) Again,

$$T_F = \frac{9}{5}T_C + 32 = \frac{9}{5}(37.0) + 32 = \boxed{98.6°\text{F}}$$

$$T = T_C + 273.15 = (37.0 + 273.15)\text{ K} = \boxed{310\text{ K}}$$

P19.4 (a) The relationship between the Kelvin and Celsius scales is given by Equation 19.1:

$$T = T_C + 273.15$$

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Thus 20.3 K converts to

$$T_C = T - 273.15 = 20.3\text{ K} - 273.15\text{ K} = \boxed{-253°\text{C}}$$

(b) The relationship between the Celsius and Fahrenheit scales is, from Equation 19.2,

$$T_F = \frac{9}{5}T_C + 32°\text{F}$$

Thus –253°C converts to

$$T_F = \frac{9}{5}T_C + 32°\text{F} = \frac{9}{5}(-253°\text{C}) + 32°\text{F} = \boxed{-423°\text{F}}$$

P19.5 (a) By Equation 19.2,

$$T_F = \frac{9}{5}T_C + 32.0°\text{F} = \frac{9}{5}(-195.81°\text{C}) + 32.0 = \boxed{-320°\text{F}}$$

(b) Applying Equation 19.1,

$$T = T_C + 273.15 = -195.81°\text{C} + 273.15 = \boxed{77.3\text{ K}}$$

***P19.6** (a) To convert from Fahrenheit to Celsius, we use

$$T_C = \frac{5}{9}(T_F - 32.0)$$

The temperature at Furnace Creek Ranch in Death Valley is

$$T_C = \frac{5}{9}(T_F - 32.0) = \frac{5}{9}(134°\text{F} - 32.0) = \boxed{56.7°\text{C}}$$

and the temperature at Prospect Creek Camp in Alaska is

$$T_C = \frac{5}{9}(T_F - 32.0) = \frac{5}{9}(-79.8°\text{F} - 32.0) = \boxed{-62.1°\text{C}}$$

(b) We find the Kelvin temperature from Equation 19.1, $T = T_C + 273.15$. The record temperature on the Kelvin scale at Furnace Creek Ranch in Death Valley is

$$T = T_C + 273.15 = 56.7°\text{C} + 273.15 = \boxed{330\text{ K}}$$

and the temperature at Prospect Creek Camp in Alaska is

$$T = T_C + 273.15 = -62.11°\text{C} + 273.15 = \boxed{211\text{ K}}$$

P19.7 Since we have a linear graph, we know that the pressure is related to the temperature as $P = A + BT_C$, where A and B are constants. To find A and B, we use the given data:

$$0.900\text{ atm} = A + B(-78.5°\text{C}) \qquad \textbf{[1]}$$

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and

$$1.635 \text{ atm} = A + B(78.0°\text{C}) \qquad \textbf{[2]}$$

Solving Equations [1] and [2] simultaneously, we find:

$$A = 1.27 \text{ atm} \quad \text{and} \quad B = 4.70 \times 10^{-3} \text{ atm/°C}$$

Therefore,

$$P = 1.27 \text{ atm} + \left(4.70 \times 10^{-3} \text{ atm/°C}\right) T_C$$

(a) At absolute zero the gas exerts zero pressure ($P = 0$), so

$$T_C = \frac{-1.27 \text{ atm}}{4.70 \times 10^{-3} \text{ atm/°C}} = \boxed{-270°\text{C}}$$

(b) At the freezing point of water, $T_C = 0$ and

$$P = 1.27 \text{ atm} + 0 = \boxed{1.27 \text{ atm}}$$

At the boiling point of water, $T_C = 100°\text{C}$, so

$$P = 1.27 \text{ atm} + \left(4.70 \times 10^{-3} \text{ atm/°C}\right)(100°\text{C}) = \boxed{1.74 \text{ atm}}$$

Section 19.4 Thermal Expansion of Solids and Liquids

P19.8 Each section can expand into the joint space to the north of it. We need think of only one section expanding. Using Equation 19.4,

$$\Delta L = L_i \alpha \Delta T = (25.0 \text{ m})\left[12.0 \times 10^{-6} \text{ (°C)}^{-1}\right](40.0°\text{C})$$

$$= \boxed{1.20 \text{ cm}}$$

: (a) By Equation 19.4,

$$\Delta L = \alpha L_i \Delta T = \left[9.00 \times 10^{-6} (°\text{C})^{-1}\right](30.0 \text{ cm})(65.0°\text{C})$$

$$= \boxed{0.176 \text{ mm}}$$

(b) The diameter is a linear dimension, so Equation 19.4 still applies:

$$\Delta L = \alpha L_i \Delta T = \left[9.00 \times 10^{-6} \text{ (°C)}^{-1}\right](1.50 \text{ cm})(65.0°\text{C})$$

$$= 8.78 \times 10^{-4} \text{ cm} = \boxed{8.78 \ \mu\text{m}}$$

(c) Using the volumetric coefficient of expansion β, and $V_i = \pi d^2 L / 4$,

$$\Delta V = \beta V_i \Delta T \approx 3\alpha V \Delta T$$

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$$\Delta V = \beta V_i \Delta T \approx 3\alpha V_i \Delta T$$

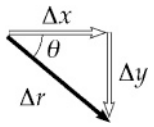
$$= 3\left[9.00\times10^{-6}\ (°\text{C})^{-1}\right]\left(\frac{30.0(\pi)(1.50)^2}{4}\ \text{cm}^3\right)(65.0°\text{C})$$

$$= \boxed{0.093\,0\ \text{cm}^3}$$

P19.10 The horizontal section expands according to $\Delta L = \alpha L_i \Delta T$.

$$\Delta x = \left[17\times10^{-6}\ (°\text{C})^{-1}\right](28.0\ \text{cm})(46.5°\text{C} - 18.0°\text{C})$$

$$= 1.36\times10^{-2}\ \text{cm}$$



ANS. FIG. P19.10

The vertical section expands similarly by

$$\Delta y = \left[17\times10^{-6}(°\text{C})^{-1}\right](134\ \text{cm})(28.5°\text{C}) = 6.49\times10^{-2}\ \text{cm}$$

The vector displacement of the pipe elbow has magnitude

$$\Delta r = \sqrt{\Delta x^2 + \Delta y^2} = \sqrt{(0.136\ \text{mm})^2 + (0.649\ \text{mm})^2} = 0.663\ \text{mm}$$

and is directed to the right below the horizontal at angle

$$\theta = \tan^{-1}\left(\frac{\Delta y}{\Delta x}\right) = \tan^{-1}\left(\frac{0.649\ \text{mm}}{0.136\ \text{mm}}\right) = 78.2°$$

$$\boxed{\Delta r = 0.663\ \text{mm to the right at } 78.2° \text{ below the horizontal}}$$

P19.11 The wire is 35.0 m long when $T_C = -20.0°\text{C}$.

$$\Delta L = L_i \bar{\alpha}\left(T - T_i\right)$$

Since $\bar{\alpha} = \alpha(20.0°\text{C}) = 1.70\times10^{-5}\ (°\text{C})^{-1}$ for Cu,

$$\Delta L = (35.0\ \text{m})\left[1.70\times10^{-5}\ (°\text{C})^{-1}\right][35.0°\text{C} - (-20.0°\text{C})]$$

$$= \boxed{+3.27\ \text{cm}}$$

***P19.12** For the dimensions to increase, $\Delta L = \alpha L_i \Delta T$:

$$1.00\times10^{-2}\ \text{cm} = \left[1.30\times10^{-4}\ (°\text{C})^{-1}\right](2.20\ \text{cm})(T - 20.0°\text{C})$$

$$T = \boxed{55.0°\text{C}}$$

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***P19.13** By Equation 19.4,

$$\Delta L = \alpha L_i \Delta T = \left[11 \times 10^{-6}\ (°\text{C})^{-1}\right](1\,300\ \text{km})[35°\text{C} - (-73°\text{C})]$$

$$= \boxed{1.54\ \text{km}}$$

The expansion can be compensated for by mounting the pipeline on rollers and placing Ω-shaped loops between straight sections. They bend as the steel changes length.

***P19.14** By Equation 19.4,

$$\Delta L = \alpha L_i \Delta T = \left[22 \times 10^{-6}\ (°\text{C})^{-1}\right](2.40\ \text{cm})(30.0°\text{C})$$

$$= \boxed{1.58 \times 10^{-3}\ \text{cm}}$$

***P19.15** (a) Following the logic in the textbook for obtaining Equation 19.6 from Equation 19.4, we can express an expansion in area as

$$\Delta A = 2\alpha A_i \Delta T$$

$$= 2\left[17.0 \times 10^{-6}\,(°\text{C})^{-1}\right](0.080\,0\ \text{m})^2\,(50.0°\text{C})$$

$$= 1.09 \times 10^{-5}\ \text{m}^2 = \boxed{0.109\ \text{cm}^2}$$

(b) The length of each side of the hole has increased. Thus, this represents an $\boxed{\text{increase}}$ in the area of the hole.

***P19.16** By Equation 19.6,

$$\Delta V = (\beta - 3\alpha) V_i \Delta T$$

$$= \left[5.81 \times 10^{-4}\,(°\text{C})^{-1} - 3\left(11.0 \times 10^{-6}\,(°\text{C})^{-1}\right)\right]$$

$$\times (50.0\ \text{gal})(20.0°\text{C})$$

$$= \boxed{0.548\ \text{gal}}$$

***P19.17** (a) By Equation 19.4, $L = L_i(1 + \alpha\Delta T)$, and

$$5.050\ \text{cm} = 5.000\ \text{cm}\left[1 + \left(24.0 \times 10^{-6}\,(°\text{C})^{-1}\right)(T - 20.0°\text{C})\right]$$

which gives $T = \boxed{437°\text{C}}$

(b) We must get $L_{\text{Al}} = L_{\text{Brass}}$ for some ΔT, or

$$L_{i,\ \text{Al}}\left(1 + \alpha_{\text{Al}}\Delta T\right) = L_{i,\ \text{Brass}}\left(1 + \alpha_{\text{Brass}}\Delta T\right)$$

$$5.000\ \text{cm}\left[1 + \left(24.0 \times 10^{-6}\,(°\text{C})^{-1}\right)\Delta T\right]$$

$$= 5.050\ \text{cm}\left[1 + \left(19.0 \times 10^{-6}\,(°\text{C})^{-1}\right)\Delta T\right]$$

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Solving for ΔT,

$$\Delta T = 2\ 080°\text{C}$$

so $\boxed{T = 2.1 \times 10^3 °\text{C}}$

(c) No. Aluminum melts at 660°C (Table 17.2). Also, although it is not in Table 17.2, internet research shows that brass (an alloy of copper and zinc) melts at about 900°C.

P19.18 We solve for the temperature T at which the brass ring would fit over the aluminum cylinder.

$$L_{\text{Al}}\left(1 + \alpha_{\text{Al}}\Delta T\right) = L_{\text{Brass}}\left(1 + \alpha_{\text{Brass}}\Delta T\right)$$

$$\Delta T = T - T_i = \frac{L_{\text{Al}} - L_{\text{Brass}}}{L_{\text{Brass}}\alpha_{\text{Brass}} - L_{\text{Al}}\alpha_{\text{Al}}}$$

$$\Delta T = \frac{10.02\text{ cm} - 10.00\text{ cm}}{(10.00\text{ cm})\left(19.0 \times 10^{-6}\ (°\text{C})^{-1}\right) - (10.02\text{ cm})\left(24.0 \times 10^{-6}\ (°\text{C})^{-1}\right)}$$

$$\Delta T = -396 = T - 20.0 \quad \rightarrow \quad T = -376°\text{C}$$

The situation is impossible because the
$\boxed{\text{required } T = -376°\text{C is below absolute zero.}}$

P19.19 (a) The original volume of the acetone we take as precisely 100 mL. After it is finally cooled to 20.0°C, its volume is

$$V_f = V_i(1 + \beta\Delta T) = (100\text{ ml})\left\{1 + \left[1.50 \times 10^{-4}(°\text{C})^{-1}\right](-15.0°\text{C})\right\}$$

$$= \boxed{99.8\text{ mL}}$$

(b) Initially, the volume of the acetone reaches the 100-mL mark on the flask, but the acetone cools and the flask warms to a temperature of 32.0 °C. Thus, the volume of the acetone decreases and the volume of the flask increases. This means the acetone will be below the 100-mL mark on the flask.

P19.20 (a) The material would expand by $\Delta L = \alpha L_i \Delta T$, or $\frac{\Delta L}{L_i} = \alpha\Delta T$, but instead feels stress

$$\frac{F}{A} = \frac{Y\Delta L}{L_i}$$

$$= Y\alpha\Delta T = \left(7.00 \times 10^9\ \text{N/m}^2\right)\left[12.0 \times 10^{-6}(\text{C}°)^{-1}\right](30.0°\text{C})$$

$$= \boxed{2.52 \times 10^6\ \text{N/m}^2}$$

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(b) The stress is less than the compressive strength, so
the concrete will not fracture.

P19.21 (a) The amount of turpentine that overflows equals the difference in the change in volume of the cylinder and the turpentine:

$$\Delta V = V_t\beta_t\Delta T - V_{\text{Al}}\beta_{\text{Al}}\Delta T = \left(\beta_t - 3\alpha_{\text{Al}}\right)V_i\Delta T$$

$$= \left[9.00\times10^{-4}\,(^\circ\text{C})^{-1} - 3\left(24.0\times10^{-6}\,(^\circ\text{C})^{-1}\right)\right] \times\left(2\ 000\ \text{cm}^3\right)(60.0^\circ\text{C})$$

$\Delta V =$ 99.4 cm^3 overflows.

(b) Find the volume of the turpentine remaining in the cylinder at 80.0°C, which is the same as the volume of the aluminum cylinder at 80.0°C:

$$V_t = V_{\text{Al}} = V_{\text{Al}i} + \beta_{\text{Al}}V_{\text{Al}i}\Delta T = V_{\text{Al}i} + 3\alpha_{\text{Al}}V_{\text{Al}i}\Delta T$$

$$= V_{\text{Al}i}\left(1+3\alpha_{\text{Al}}\Delta T\right)$$

$$= \left(2\ 000\ \text{cm}^3\right)\left[1+3\left(24\times10^{-6}\,(^\circ\text{C})^{-1}\right)(60.0^\circ\text{C})\right]$$

$$= 2\ 008.64\ \text{cm}^3 =$$ 2.01 L

(c) Find the volume of the turpentine in the cylinder after it cools back to 20.0°C:

$$V = V_{ti} + \beta_tV_{ti}\Delta T = V_{ti}\left(1 + \beta_t\Delta T\right)$$

$$= \left(2\ 008.64\ \text{cm}^3\right)\left[1 + \left(9\times10^{-4}\,(^\circ\text{C})^{-1}\right)(-60.0^\circ\text{C})\right]$$

$$= 1\ 900.17\ \text{cm}^3$$

Find the percentage of the cylinder that is empty at 20.0°C:

$$\frac{2\ 000\ \text{cm}^3 - 1\ 900.17\ \text{cm}^3}{2\ 000\ \text{cm}^3} = 4.99\%$$

Find the empty height of the cylinder above the turpentine:

$$(4.99\%)(20.0\ \text{cm}) =$$ 0.998 cm

P19.22 We model the wire as contracting according to $\Delta L = \alpha L_i\Delta T$ and then stretching according to

$$\text{stress} = \frac{F}{A} = Y\frac{\Delta L}{L_i} = \frac{Y}{L_i}\alpha L_i\Delta T = Y\alpha\Delta T$$

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(a) We find the tension from

$$F = YA\alpha\Delta T$$
$$= \left(20.0\times10^{10}\ \text{N/m}^2\right)\left(4.00\times10^{-6}\ \text{m}^2\right)$$
$$\times\left[11\times10^{-6}\ (\text{°C})^{-1}\right](45.0\text{°C})$$
$$= \boxed{396\ \text{N}}$$

(b) $$\Delta T = \frac{\text{stress}}{Y\alpha} = \frac{3.00\times10^{8}\ \text{N/m}^2}{\left(20.0\times10^{10}\ \text{N/m}^2\right)\left(11\times10^{-6}/\text{C°}\right)} = 136\text{°C}$$

To increase the stress the temperature must decrease to $35\text{°C} - 136\text{°C} = \boxed{-101\text{°C}}$.

(c) $\boxed{\text{The original length divides out, so the answers would not change.}}$

P19.23 (a) The density of a sample of lead of mass m = 20.0 kg, volume V_0, at temperature T_0 is

$$\rho_0 = \frac{m}{V_0} = 11.3\times10^{3}\ \text{kg/m}^3$$

For a temperature change $\Delta T = T - T_0$, the same mass m occupies a larger volume $V = V_0(1+\beta\Delta T)$; therefore, the density is

$$\rho = \frac{m}{V_0(1+\beta\Delta T)} = \frac{\rho_0}{(1+\beta\Delta T)}$$

where $\beta = 3\alpha$, and $\alpha = 29\times10^{-6}\ (\text{°C})^{-1}$.

For a temperature change of from 0.00°C to 90.0°C,

$$\rho = \frac{\rho_0}{(1+\beta\Delta T)} = \frac{11.3\times10^{3}\ \text{kg/m}^3}{1+3\left(29\times10^{-6}\ (\text{°C})^{-1}\right)(90.0\ \text{°C})}$$
$$= \boxed{11.2\times10^{3}\ \text{kg/m}^3}$$

(b) The mass is still the same, $\boxed{20.0\ \text{kg}}$, because a temperature change would not change the mass.

P19.24 (a) The density of a solid substance of mass m, volume V_0, at temperature T_0 is

$$\rho_0 = \frac{m}{V_0}$$

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For a temperature change $\Delta T = T - T_0$, the same mass m occupies a larger volume $V = V_0(1+\beta\Delta T)$; therefore, the density is

$$\rho = \frac{m}{V_0(1+\beta\Delta T)} = \boxed{\frac{\rho_0}{1+\beta\Delta T}}$$

(b) The mass is still the same, $\boxed{m}$, because a temperature change would not change the mass.

P19.25 From Equation 19.3, the difference in Celsius temperature in the underground tank and the tanker truck is

$$\Delta T_C = \frac{5}{9}(\Delta T_F) = \frac{5}{9}(95.0°\text{F} - 52.0°\text{F}) = 23.9°\text{C}$$

If $V_{52.0°\text{F}}$ is the volume of gasoline that fills the tank at 52.0°F, the volume this quantity of gas would occupy on the tanker truck at 95.0°F is

$$\begin{aligned} V_{95.0°\text{F}} &= V_{52.0°\text{F}} + \Delta V = V_{52.0°\text{F}} + \beta V_{52.0°\text{F}}\Delta T = V_{52.0°\text{F}}(1+\beta\Delta T) \\ &= (1.00\times10^3\ \text{gal})\left\{1+\left[9.6\times10^{-4}\ (°\text{C})^{-1}\right](23.9°\text{C})\right\} \\ &= 1.02\times10^3\ \text{gal} \end{aligned}$$

Section 19.5 Macroscopic Description of an Ideal Gas

P19.26 If the volume and the temperature are both constant, the ideal gas law gives

$$\frac{P_f \cancel{V_f}}{P_i \cancel{V_i}} = \frac{n_f \cancel{RT_f}}{n_i \cancel{RT_i}}$$

or $$n_f = \left(\frac{P_f}{P_i}\right)n_i = \left(\frac{5.00\ \text{atm}}{25.0\ \text{atm}}\right)(1.50\ \text{mol}) = 0.300\ \text{mol}$$

so the amount of gas to be withdrawn is

$$\Delta n = n_i - n_f = 1.50\ \text{mol} - 0.300\ \text{mol} = \boxed{1.20\ \text{mol}}$$

P19.27 The initial and final absolute temperatures are

$$T_i = T_{C,i} + 273 = (25.0+273)\ \text{K} = 298\ \text{K}$$

and

$$T_f = T_{C,f} + 273 = (75.0+273)\ \text{K} = 348\ \text{K}$$

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The volume of the tank is assumed to be unchanged, or $V_f = V_i$. Also, since two-thirds of the gas is withdrawn, $n_f = n_i/3$. Thus, from the ideal gas law, we obtain

$$\frac{P_f \cancel{V_f}}{P_i \cancel{V_i}} = \frac{n_f \cancel{R} T_f}{n_i \cancel{R} T_i}$$

or

$$P_f = \left(\frac{n_f}{n_i}\right)\left(\frac{T_f}{T_i}\right)P_i = \left(\frac{1}{3}\right)\left(\frac{348\text{ K}}{298\text{ K}}\right)(11.0\text{ atm}) = \boxed{4.28\text{ atm}}$$

P19.28 When the tank has been prepared and is ready to use it contains 1.00 L of air and 4.00 L of water. Consider the air in the tank during one discharge process. We suppose that the process is slow enough that the temperature remains constant. Then as the pressure drops from 2.40 atm to 1.20 atm, the volume of the air doubles ($PV \approx$ constant) resulting in 1.00 L of water expelled and 3.00 L remaining. In the second discharge, the air volume doubles from 2.00 L to 4.00 L and 2.00 L of water is sprayed out. In the third discharge, only the last 1.00 L of water comes out.

In each pump-up-and-discharge cycle, the volume of air in the tank doubles. Thus 1.00 L of water is driven out by the air injected at the first pumping, 2.00 L by the second, and only the remaining 1.00 L by the third. Each person could more efficiently use his device by starting with the tank half full of water, instead of 80% full.

P19.29 (a) From the ideal gas law,

$$n = \frac{PV}{RT} = \frac{(9.00\text{ atm})\left(1.013\times10^{5}\text{ Pa/atm}\right)\left(8.00\times10^{-3}\text{ m}^3\right)}{(8.314\text{ N}\cdot\text{mol K})(293\text{ K})}$$

$$= \boxed{2.99\text{ mol}}$$

(b) The number of molecules is

$$N = nN_A = (2.99\text{ mol})\left(6.02\times10^{23}\text{ molecules/mol}\right)$$

$$= \boxed{1.80\times10^{24}\text{ molecules}}$$

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P19.30 (a) From $PV = nRT$, we obtain $n = \dfrac{PV}{RT}$. Then

$$m = nM = \frac{PVM}{RT}$$

$$= \frac{\left(1.013\times10^{5}\ \text{Pa}\right)(0.100\ \text{m})^{3}\left(28.9\times10^{-3}\ \text{kg/mol}\right)}{\left(8.314\ \text{J/mol}\cdot\text{K}\right)(300\ \text{K})}$$

$$= \boxed{1.17\times10^{-3}\ \text{kg}}$$

(b) $F_g = mg = \left(1.17\times10^{-3}\ \text{kg}\right)\left(9.80\ \text{m/s}^2\right) = \boxed{11.5\ \text{mN}}$

(c) $F = PA = \left(1.013\times10^{5}\ \text{N/m}^2\right)(0.100\ \text{m})^2 = \boxed{1.01\ \text{kN}}$

(d) The $\boxed{\text{molecules must be moving very fast}}$ to hit the walls hard.

P19.31 The equation of state of an ideal gas is $PV = nRT$, so we need to solve for the number of moles to find N.

$$n = \frac{PV}{RT} = \frac{\left(1.01\times10^{5}\ \text{N/m}^2\right)\left[(10.0\ \text{m})(20.0\ \text{m})(30.0\ \text{m})\right]}{\left(8.314\ \text{J/mol}\cdot\text{K}\right)(293\ \text{K})}$$

$$= 2.49\times10^{5}\ \text{mol}$$

Then,

$$N = nN_A = \left(2.49\times10^{5}\ \text{mol}\right)\left(6.022\times10^{23}\ \text{molecules/mol}\right)$$

$$= \boxed{1.50\times10^{29}\ \text{molecules}}$$

P19.32 From the ideal gas law, $PV = nRT$, and

$$\frac{m_f}{m_i} = \frac{n_f}{n_i} = \frac{P_f V_f}{RT_f}\frac{RT_i}{P_i V_i} = \frac{P_f}{P_i}$$

so $$m_f = m_i\left(\frac{P_f}{P_i}\right)$$

and $$|\Delta m| = m_i - m_f = m_i\left(\frac{P_i - P_f}{P_i}\right) = 12.0\ \text{kg}\left(\frac{41.0\ \text{atm} - 26.0\ \text{atm}}{41.0\ \text{atm}}\right)$$

$$= \boxed{4.39\ \text{kg}}$$

P19.33 (a) From the ideal gas law, $PV = nRT$, so

$$n = \frac{PV}{RT} = \frac{\left(1.013\times10^{5}\ \text{Pa}\right)\left(1.00\ \text{m}^3\right)}{\left(8.314\ \text{J/mol}\cdot\text{K}\right)(293\ \text{K})} = \boxed{41.6\ \text{mol}}$$

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(b) $m = nM = (41.6 \text{ mol})(28.9 \text{ g/mol}) = \boxed{1.20 \text{ kg}}$

(c) $\boxed{\text{This value agrees with the tabulated density}}$ of 1.20 kg/m^3 at 20.0°C.

***P19.34** One mole of helium contains Avogadro's number of molecules and has a mass of 4.00 g. Let us call m_0 the mass of one atom, and we have

$$N_A m_0 = 4.00 \text{ g/mol}$$

or

$$m_0 = \frac{4.00 \text{ g/mol}}{6.02 \times 10^{23} \text{ molecules/mol}} = 6.64 \times 10^{-24} \text{ g/molecule}$$

$$= \boxed{6.64 \times 10^{-27} \text{ kg}}$$

***P19.35** The CO_2 is far from liquefaction, so after it comes out of solution it behaves as an ideal gas. Its molar mass is M = 12.0 g/mol + 2(16.0 g/mol) = 44.0 g/mol. The quantity of gas in the cylinder is

$$n = \frac{m_{\text{sample}}}{M} = \frac{6.50 \text{ g}}{44.0 \text{ g/mol}} = 0.148 \text{ mol}$$

Then $PV = nRT$ gives

$$V = \frac{nRT}{P}$$

$$= \frac{0.148 \text{ mol}(8.314 \text{ J/mol} \cdot \text{K})(273.15 \text{ K} + 20°\text{C})}{1.013 \times 10^5 \text{ N/m}^2} \times \left(\frac{1 \text{ N} \cdot \text{m}}{1 \text{ J}}\right)\left(\frac{10^3 \text{ L}}{1 \text{ m}^3}\right)$$

$$= \boxed{3.55 \text{ L}}$$

P19.36 We use Equation 19.10, $PV = Nk_BT$:

$$N = \frac{PV}{k_B T} = \frac{(1.00 \times 10^{-9} \text{ Pa})(1.00 \text{ m}^3)}{(1.38 \times 10^{-23} \text{ J/K})(300 \text{ K})} = \boxed{2.42 \times 10^{11} \text{ molecules}}$$

P19.37 (a) Initially, $P_i V_i = n_i R T_i$: $(1.00 \text{ atm}) V_i = n_i R[(10.0°\text{C} + 273.15) \text{ K}]$ **[1]**

Finally, $P_f V_f = n_f R T_f$: $P_f (0.280 V_i) = n_i R[(40.0°\text{C} + 273.15) \text{ K}]$ **[2]**

Dividing [2] by [1]: $\frac{0.280 P_f}{1.00 \text{ atm}} = \frac{313.15 \text{ K}}{283.15 \text{ K}}$

giving $P_f =$ $\boxed{3.95 \text{ atm} = 4.00 \times 10^5 \text{ Pa}}$

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(b) After being driven, $P_d(1.02)(0.280V_i) = n_i R(85.0°\text{C} + 273.15)\ \text{K}$ **[3]**

Dividing [3] by [1]: $\dfrac{(1.02)(0.280)P_d}{1.00\ \text{atm}} = \dfrac{358.15\ \text{K}}{283.15\ \text{K}}$

$$P_d = 4.43\ \text{atm} = \boxed{4.49 \times 10^5\ \text{Pa}}$$

P19.38 The air in the tube is far from liquefaction, so it behaves as an ideal gas. At the ocean surface it is described by $P_t V_t = nRT$, where P_t = 1 atm, $V_t = A$ (6.50 cm), and A is the cross-sectional area of the interior of the tube. At the bottom of the dive,

$$P_b V_b = nRT = P_b A(6.50\ \text{cm} - 2.70\ \text{cm})$$

By division,

$$\frac{P_b(3.80\ \text{cm})}{(1\ \text{atm})(6.50\ \text{cm})} = 1$$

$$P_b = \left(1.013 \times 10^5\ \text{N/m}^2\right)\left(\frac{6.50\ \text{cm}}{3.80\ \text{cm}}\right) = 1.73 \times 10^5\ \text{N/m}^2$$

The salt water enters the tube until the air pressure is equal to the water pressure at depth, which is described by

$$P_b = P_t + \rho g h$$

$$1.73 \times 10^5\ \text{N/m}^2 = 1.013 \times 10^5\ \text{N/m}^2 + \left(1\ 030\ \text{kg/m}^3\right)\left(9.80\ \text{m/s}^2\right)h$$

solving for the depth h of the dive gives

$$h = \frac{7.20 \times 10^4\ \text{kg} \cdot \text{m} \cdot \text{m}^2 \cdot \text{s}^2}{1.01 \times 10^4\ \text{s}^2 \cdot \text{m}^2 \cdot \text{kg}} = \boxed{7.13\ \text{m}}$$

P19.39 The density of the air inside the balloon, ρ_{in}, must be reduced until the buoyant force of the outside air is at least equal to the weight of the balloon plus the weight of the air inside it:

$$\sum F_y = 0: \quad B - W_{\text{air inside}} - W_{\text{balloon}} = 0$$

$$\rho_{\text{out}} g V - \rho_{\text{in}} g V - m_b g = 0 \quad \rightarrow \quad (\rho_{\text{out}} - \rho_{\text{in}})V = m_b$$

where $\rho_{\text{out}} = 1.244\ \text{kg/m}^3$, $V = 400\ \text{m}^3$, and $m_b = 200\ \text{kg}$.

From $PV = nRT$, $\dfrac{n}{V} = \dfrac{P}{RT}$. This equation means that at constant pressure the density is inversely proportional to the temperature. Thus, the density of the hot air inside the balloon is

$$\rho_{\text{in}} = \rho_{\text{out}}\left(\frac{283\ \text{K}}{T_{\text{in}}}\right)$$

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Substituting this result into the condition $(\rho_{\text{out}} - \rho_{\text{in}})V = m_b$ gives

$$\rho_{\text{out}}\left(1 - \frac{283\text{ K}}{T_{in}}\right) = \frac{m_b}{V} \quad \rightarrow \quad \frac{283\text{ K}}{T_{in}} = 1 - \frac{m_b}{\rho_{out}V}$$

$$\rightarrow \quad T_{in} = \frac{283\text{ K}}{\left(1 - \dfrac{m_b}{\rho_{out}V}\right)}$$

$$T_{in} = \frac{283\text{ K}}{\left(1 - \dfrac{200\text{ kg}}{(1.244\text{ kg/m}^3)(400\text{ m}^3)}\right)} = \boxed{473\text{ K}}$$

***P19.40** To compute the mass of air leaving the room, we begin with the ideal gas law:

$$P_0V = n_1RT_1 = \left(\frac{m_1}{M}\right)RT_1$$

As the temperature is increased at constant pressure,

$$P_0V = n_2RT_2 = \left(\frac{m_2}{M}\right)RT_2$$

Subtracting the two equations gives

$$\boxed{m_1 - m_2 = \frac{P_0VM}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)}$$

P19.41 At depth, $P = P_0 + \rho gh$ and $PV_i = nRT_i$

At the surface, $P_0V_f = nRT_f$: $\quad \dfrac{P_0V_f}{(P_0 + \rho gh)V_i} = \dfrac{T_f}{T_i}$

Therefore, $V_f = V_i\left(\dfrac{T_f}{T_i}\right)\left(\dfrac{P_0 + \rho gh}{P_0}\right)$ and

$$V_f = 1.00\text{ cm}^3\left(\frac{293\text{ K}}{278\text{ K}}\right) \times \left(\frac{(1.013\times10^5\text{ Pa}) + (1\,030\text{ kg/m}^3)(9.80\text{ m/s}^2)(25.0\text{ m})}{1.013\times10^5\text{ Pa}}\right)$$

$$V_f = \boxed{3.68\text{ cm}^3}$$

P19.42 My bedroom is 4 m long, 4 m wide, and 2.4 m high, enclosing air at 100 kPa and 20°C = 293 K. Think of the air as 80.0% N_2 and 20.0% O_2.

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Avogadro's number of molecules has mass

$$(0.800)(28.0\ \text{g/mol})+(0.200)(32.0\ \text{g/mol})=0.028\ 8\ \text{kg/mol}$$

Then $PV = nRT = \left(\frac{m}{M}\right)RT$ gives

$$m=\frac{PVM}{RT}=\frac{(1.00\times10^5\ \text{N/m}^2)(38.4\ \text{m}^3)(0.028\ 8\ \text{kg/mol})}{(8.314\ \text{J/mol}\cdot\text{K})(293\ \text{K})}$$

$$=45.4\ \text{kg}\ \boxed{\sim10^2\ \text{kg}}$$

P19.43 Pressure inside the cooker is due to the pressure of water vapor plus the air trapped inside. The pressure of the water vapor is

$$P_\text{v}=\frac{nRT}{V}=\left(\frac{9.00\ \text{g}}{18.0\ \text{g/mol}}\right)\left(\frac{8.314\ \text{J}}{\text{mol K}}\right)\left(\frac{773\ \text{K}}{2.00\times10^{-3}\ \text{m}^3}\right)$$

$$=1.61\ \text{MPa}$$

We find the pressure of the air at constant volume, assuming the initial temperature is 10°C:

$$\frac{P_{a2}}{P_{a1}}=\frac{T_2}{T_1}\quad\rightarrow\quad P_{a2}=P_{a1}\frac{T_2}{T_1}=(101\ \text{kPa})\frac{773\ \text{K}}{283\ \text{K}}$$

$$=276\ \text{kPa}=0.276\ \text{MPa}$$

The total pressure is

$$P=P_\text{v}+P_{a2}=1.61\ \text{MPa}\ +0.276\ \text{MPa}=\boxed{1.89\ \text{MPa}}$$

P19.44 If P_{gi} is the initial gauge pressure of the gas in the cylinder, the initial absolute pressure is $P_{\text{i,abs}} = P_{gi} + P_0$, where P_0 is the exterior pressure. Likewise, the final absolute pressure in the cylinder is $P_{\text{f,abs}} = P_{gf} + P_0$, where P_{gf} is the final gauge pressure. The initial and final masses of gas in the cylinder are $m_i = n_i M$ and $m_f = n_f M$, where n is the number of moles of gas present and M is the molecular weight of this gas. Thus, $m_f/m_i = n_f/n_i$.

We assume the cylinder is a rigid container whose volume does not vary with internal pressure. Also, since the temperature of the cylinder is constant, its volume does not expand or contract. Then, the ideal gas law (using absolute pressures) with both temperature and volume constant gives

$$\frac{P_{\text{f,abs}}\cancel{V}}{P_{\text{i,abs}}\cancel{V}}=\frac{n_f\cancel{RT}}{n_i\cancel{RT}}=\frac{m_f}{m_i}\qquad\text{or}\qquad m_f=m_i\left(\frac{P_{\text{f,abs}}}{P_{\text{i,abs}}}\right)$$

and in terms of gauge pressures,

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$$\boxed{m_f = m_i\left(\frac{P_{gf} + P_0}{P_{gi} + P_0}\right)}$$

Additional Problems

P19.45 The astronauts exhale this much CO_2:

$$n = \frac{m_{\text{sample}}}{M}$$
$$= \left(\frac{1.09\text{ kg}}{\text{astronaut}\cdot\text{day}}\right)\left(\frac{1\,000\text{ g}}{1\text{ kg}}\right)$$
$$\times(3\text{ astronauts})(7\text{ days})\left(\frac{1\text{ mol}}{44.0\text{ g}}\right) = 520\text{ mol}$$

Then 520 mol of methane is generated. It is far from liquefaction and behaves as an ideal gas, so the pressure is

$$P = \frac{nRT}{V} = \frac{(520\text{ mol})(8.314\text{ J/mol}\cdot\text{K})(273.15\text{ K} - 45\text{ K})}{150\times10^{-3}\text{ m}^3}$$
$$= \boxed{6.57\times10^6\text{ Pa}}$$

P19.46 We must first convert both the initial and final temperatures to Celsius:

$$T_C = \frac{5}{9}(T_F - 32)$$

Thus, $$T_{initial} = \frac{5}{9}(T_{F,initial} - 32) = \frac{5}{9}(15.000 - 32.000) = -9.444^\circ\text{C}$$

$$T_{final} = \frac{5}{9}(T_{F,final} - 32) = \frac{5}{9}(90.000 - 32.000) = 32.222^\circ\text{C}$$

The length of the steel beam after heating is L_f, and the linear expansion of the beam follows the equation: $\Delta L = L_f - L_i = \alpha L_i \Delta T$

Thus,

$$L_f = \alpha L_i(T_f - T_i) + L_i$$
$$= (11\times10^{-6}{}^\circ\text{C}^{-1})(35.000\text{ m})[32.222^\circ\text{C} - (-9.444^\circ\text{C})]$$
$$+ 35.000\text{ m}$$
$$= 0.016\text{ m} + 35.000\text{ m} = \boxed{35.016\text{ m}}$$

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P19.47 (a) The diameter is a linear dimension, so we consider the linear expansion of steel:

$$\begin{aligned} d &= d_0\left[1+\alpha(\Delta T)\right] \\ &= (2.540\text{ cm})\left[1+\left(11\times10^{-6}\ (^\circ\text{C})^{-1}\right)(100.00^\circ\text{C}-25.00^\circ\text{C})\right] \\ &= \boxed{2.542\text{ cm}} \end{aligned}$$

(b) If the volume increases by 1%, then $\Delta V = (1.000\times10^{-2})\,V_0$. Then, using $\Delta V = \beta V_0(\Delta T)$, where $\beta = 3\alpha$ is the volume expansion coefficient, we find

$$\Delta T = \frac{\Delta V/V_0}{\beta} = \frac{1.000\times10^{-2}}{3\left[11.0\times10^{-6}\ (^\circ\text{C})^{-1}\right]} = \boxed{3.0\times10^{2}\ ^\circ\text{C}}$$

P19.48 The ideal gas law will be used to find the pressure in the tire at the higher temperature. However, one must always be careful to use absolute temperatures and absolute pressures in all ideal gas law calculations.

The initial absolute pressure is

$$P_i = P_{i,\text{gauge}} + P_{\text{atm}} = 2.50\text{ atm} + 1.00\text{ atm} = 3.50\text{ atm}$$

The initial absolute temperature is

$$T_i = T_{i,\text{C}} + 273.15 = (15.0 + 273.15)\text{ K} = 288.2\text{ K}$$

and the final absolute temperature is

$$T_f = T_{f,\text{C}} + 273.15 = (45 + 273.15)\text{ K} = 318.2\text{ K}$$

The ideal gas law, with volume and quantity of gas constant, gives the final absolute pressure as

$$\frac{P_f \cancel{V_f}}{P_i \cancel{V_i}} = \frac{\cancel{n_f} R T_f}{\cancel{n_i} R T_i}$$

$$\Rightarrow \quad P_f = \left(\frac{T_f}{T_i}\right)P_i = \left(\frac{318.2\text{ K}}{288.2\text{ K}}\right)(3.50\text{ atm}) = 3.86\text{ atm}$$

The final gauge pressure in the tire is

$$P_{f,\text{gauge}} = P_f - P_{\text{atm}} = 3.86\text{ atm} - 1.00\text{ atm} = \boxed{2.86\text{ atm}}$$

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***P19.49** Some gas will pass through the porous plug from the reaction chamber 1 to the reservoir 2 as the reaction chamber is heated, but the net quantity of gas stays constant according to $n_{i1} + n_{i2} = n_{f1} + n_{f2}$.

Assuming the gas is ideal, we apply $n = \dfrac{PV}{RT}$ to each term:

$$\frac{P_i V_0}{(300\text{ K})R} + \frac{P_i (4V_0)}{(300\text{ K})R} = \frac{P_f V_0}{(673\text{ K})R} + \frac{P_f (4V_0)}{(300\text{ K})R}$$

$$1\text{ atm}\left(\frac{5}{300\text{ K}}\right) = P_f \left(\frac{1}{673\text{ K}} + \frac{4}{300\text{ K}}\right)$$

$$\boxed{P_f = 1.12\text{ atm}}$$

P19.50 Let us follow the cycle, assuming that the conditions for ideal gases apply. (That is, that the gas never comes near the conditions for which a phase transition would occur.)

We may use the ideal gas law:

$$PV = nRT$$

in which the pressure and temperature must be total pressure (in pascals or atm, depending on the units of R chosen), and absolute temperature (in K).

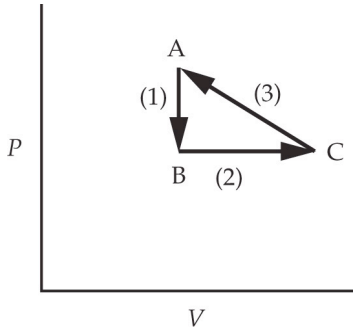
For **stage (1)** of the cycle, the process is:

$$PV = nRT \rightarrow V\Delta P = nR\Delta T$$

And, because only T and P vary:

$$\frac{\Delta T}{\Delta P} = \frac{V}{nR} = \text{const.}$$

Thus: $$\frac{T_f}{P_f} = \frac{T_i}{P_i} = \frac{V}{nR} = \text{const.}$$



ANS. FIG. P16.70

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However, when we substitute into the temperature–pressure relation for **stage (1)**, we obtain:

$$\frac{T_f}{P_f}=\frac{T_i}{P_i} \quad \rightarrow \quad T_{\text{B}}=T_f=\frac{P_f}{P_i}T_i=\frac{0.870\text{ atm}}{1.000\text{ atm}}(150°C+273.15)$$

$$=368.14\text{ K}=\boxed{95.0°\text{C}}$$

T falls below 100°C, so steam condenses and the expensive apparatus falls (assuming that the boiling point does not change significantly with the change in pressure).

P19.51 We assume the dimensions of the capillary tube do not change.

For mercury, $\beta=1.82\times10^{-4}(°\text{C})^{-1}$

and for Pyrex glass, $\alpha=3.20\times10^{-6}(°\text{C})^{-1}$

The volume of the liquid increases as $\Delta V_\ell=V\beta\Delta T$.

The volume of the shell increases as $\Delta V_g=3\alpha V\Delta T$.

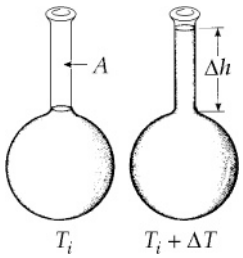
Therefore, the overflow in the capillary is $\Delta V_c=V\Delta T(\beta-3\alpha)$, and in the capillary $\Delta V_c=A\Delta h$.

$$\Delta V_c=A\Delta h=V\Delta T(\beta-3\alpha) \quad \rightarrow \quad \Delta h=\frac{(\beta-3\alpha)V\Delta T}{A}$$

$$\Delta h=\left(\frac{1}{\pi\left(\frac{0.004\ 00\times10^{-2}\text{ m}}{2}\right)^2}\right)\left[1.82\times10^{-4}(°\text{C})^{-1}-3\left(3.20\times10^{-6}(°\text{C})^{-1}\right)\right]$$

$$\times\left[\frac{4}{3}\pi\left(\frac{0.250\times10^{-2}\text{ m}}{2}\right)^3\right](30.0°\text{C})$$

$$\Delta h=3.37\times10^{-2}\text{ m}=\boxed{3.37\text{ cm}}$$



ANS. FIG. P19.51

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P19.52 We assume the dimensions of the capillary do not change. The volume of the liquid increases by $\Delta V_\ell = V\beta\Delta T$. The volume of the shell increases by $\Delta V_g = 3\alpha V\Delta T$. Therefore, the overflow in the capillary is $\Delta V_c = V\Delta T(\beta - 3\alpha)$; and in the capillary $\Delta V_c = A\Delta h$.

Therefore, $\boxed{\Delta h = (\beta - 3\alpha)\dfrac{V\Delta T}{A}}$.

P19.53 The fundamental frequency played by the cold-walled flute is

$$f_i = \frac{v}{\lambda_i} = \frac{v}{2L_i}$$

Assuming the change in the speed of sound as a function of temperature is negligible, when the instrument warms up

$$f_f = \frac{v}{\lambda_f} = \frac{v}{2L_f} = \frac{v}{2L_i(1+\alpha\Delta T)} = \frac{f_i}{1+\alpha\Delta T}$$

The final frequency is lower. The change in frequency is

$$\Delta f = f_f - f_i = f_i\left(\frac{1}{1+\alpha\Delta T} - 1\right)$$

$$\Delta f = \frac{(343\ \text{m/s})}{2(0.655\ \text{m})}\left(\frac{1}{1+(24.0\times10^{-6}/\text{C°})(15.0°\text{C})} - 1\right)$$

$$= \boxed{-0.0942\ \text{Hz}}$$

This change in frequency is imperceptibly small.

P19.54 Let L_0 represent the length of each bar at 0°C.

(a) In the diagram consider the right triangle that each invar bar makes with one half of the aluminum bar. We have

$$\sin\left(\frac{\theta}{2}\right) = \frac{L_0(1+\alpha_{\text{Al}}\Delta T)/2}{L_0} = \frac{L_0(1+\alpha_{\text{Al}}\Delta T)}{2L_0}.$$

Solving gives

$$\boxed{\theta = 2\sin^{-1}\left(\frac{1+\alpha_{\text{Al}}T_{\text{C}}}{2}\right)}$$

where T_{C} is the Celsius temperature.

(b) Yes. If the temperature drops, the negative value of Celsius temperature describes the contraction. So the answer is accurate.

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(c) Yes. At $T_C = 0$ we have $\theta = 2\sin^{-1}(1/2) = 60.0°$, and this is accurate.

(d) From the same triangle we have

$$\sin\left(\frac{\theta}{2}\right) = \frac{L_0(1+\alpha_{\text{Al}}\Delta T)}{2L_0(1+\alpha_{\text{invar}}\Delta T)}$$

giving

$$\boxed{\theta = 2\sin^{-1}\left(\frac{1+\alpha_{\text{Al}}T_C}{2(1+\alpha_{\text{invar}}T_C)}\right)}$$

(e) The greatest angle is at 660°C,

$$\theta = 2\sin^{-1}\left(\frac{1+\alpha_{\text{Al}}T_C}{2(1+\alpha_{\text{invar}}T_C)}\right) = 2\sin^{-1}\left(\frac{1+(24\times10^{-6})660}{2(1+[0.9\times10^{-6}]660)}\right)$$

$$= 2\sin^{-1}\left(\frac{1.015\ 84}{2.0011\ 88}\right) = 2\sin^{-1}0.508 = \boxed{61.0°}$$

(f) The smallest angle is at –273°C,

$$\theta = 2\sin^{-1}\left(\frac{1+(24\times10^{-6})(-273)}{2(1+[0.9\times10^{-6}][-273])}\right)$$

$$= 2\sin^{-1}\left(\frac{0.9934}{1.9995}\right) = 2\sin^{-1}0.497 = \boxed{59.6°}$$

P19.55 The excess expansion of the brass is

$$\Delta L_{\text{rod}} - \Delta L_{\text{tape}} = (\alpha_{\text{brass}} - \alpha_{\text{steel}})L_i\Delta T$$

$$\Delta(\Delta L) = (19.0 - 11.0)\times10^{-6}\ (°\text{C})^{-1}(0.950\ \text{m})(35.0°\text{C})$$

$$\Delta(\Delta L) = 2.66\times10^{-4}\ \text{m}$$

(a) The rod contracts more than the tape to a length reading

$$0.950\ 0\ \text{m} - 0.000\ 266\ \text{m} = \boxed{94.97\ \text{cm}}$$

(b) $0.950\ 0\ \text{m} + 0.000\ 266\ \text{m} = \boxed{95.03\ \text{cm}}$

P19.56 At 0°C, mass m of gasoline occupies volume $V_{0°\text{C}}$; the density of the gasoline is

$$\rho_{0°\text{C}} = \frac{m}{V_{0°\text{C}}} = 730\ \text{kg/m}^3$$

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At temperature ΔT above 0°C, the same mass of gasoline occupies a larger volume $V = V_{0°C}(1+\beta\Delta T)$: the density of the gasoline is

$\rho = \dfrac{m}{V_{0°C}(1+\beta\Delta T)} = \dfrac{\rho_{0°C}}{1+\beta\Delta T}$, which is slightly smaller than $\rho_{0°C}$.

For the same volume of gasoline, the difference in mass between gasoline at 0°C and gasoline at 20.0°C is

$$\Delta m = \rho_{0°C}V - \rho V = \rho_{0°C}V - \frac{\rho_{0°C}}{1+\beta\Delta T}V$$

$$\Delta m = \rho_{0°C}V\left(1 - \frac{1}{1+\beta\Delta T}\right)$$

$$\Delta m = \left[\left(730\ \text{kg/m}^3\right)\left(10.0\ \text{gal}\right)\left(\frac{0.003\,80\ \text{m}^3}{1.00\ \text{gal}}\right)\right] \times\left(1 - \frac{1}{1+\left(9.60\times10^{-4}(°\text{C})^{-1}\right)(20.0°\text{C})}\right)$$

$$\Delta m = \boxed{0.523\ \text{kg}}$$

P19.57 (a) $\rho = \dfrac{m}{V}$ and $d\rho = -\dfrac{m}{V^2}dV$

For very small changes in V and ρ, this can be expressed as

$$\Delta\rho = -\frac{m}{V}\frac{\Delta V}{V} = -\rho\beta\Delta T \quad \rightarrow \quad \frac{\Delta\rho}{\rho} = -\beta\Delta T$$

(b) As the temperature increases, the density decreases.

(c) For water we have $\beta = -\dfrac{\Delta\rho}{\rho\Delta T} = -\dfrac{0.999\,7\ \text{g/cm}^3 - 1.000\,0\ \text{g/cm}^3}{\left(1.000\,0\ \text{g/cm}^3\right)(10.0°\text{C} - 4.0°\text{C})}$

$$= \boxed{5\times10^{-5}(°\text{C})^{-1}}$$

(d) $\beta = -\dfrac{\Delta\rho}{\rho\Delta T} = -\dfrac{1.000\,0\ \text{g/cm}^3 - 0.999\,9\ \text{g/cm}^3}{\left(1.000\,0\ \text{g/cm}^3\right)(4.00°\text{C} - 0.00°\text{C})}$

$$= \boxed{-2.5\times10^{-5}(°\text{C})^{-1}}$$

P19.58 (a) From $PV = nRT$, the volume is $V = \left(\dfrac{nR}{P}\right)T$.

Therefore, when pressure is held constant, $\dfrac{dV}{dT} = \dfrac{nR}{P} = \dfrac{V}{T}$.

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Thus, $\beta \equiv \left(\frac{1}{V}\right)\frac{dV}{dT} = \left(\frac{1}{V}\right)\frac{V}{T}$ or $\beta = \boxed{\frac{1}{T}}$

(b) At $T = 0°\text{C} = 273.15\ \text{K}$, this predicts $\beta = \frac{1}{273\ \text{K}} = \boxed{3.66 \times 10^{-3}\ \text{K}^{-1}}$.

Experimental values are:

(c) $\boxed{\beta_{\text{He}} = 3.665 \times 10^{-3}\ \text{K}^{-1},\ \text{this agrees within 0.06\%}\ \text{of the tabulated value.}}$

(d) $\boxed{\beta_{\text{air}} = 3.67 \times 10^{-3}\ \text{K}^{-1},\ \text{this agrees within 0.2\%}\ \text{of the tabulated value.}}$

P19.59 (a) Using the expression for the period T_P of a pendulum, we have

$$T_P = 2\pi\sqrt{\frac{L}{g}} \quad \rightarrow \quad dT_P = 2\pi\sqrt{\frac{1}{g}}\left(\frac{1}{2}\right)\frac{dL}{\sqrt{L}}$$

$$= 2\pi\sqrt{\frac{L}{g}}\left(\frac{1}{2}\right)\frac{dL}{L} = T_P\left(\frac{1}{2}\right)\frac{dL}{L}$$

$$\frac{dT_P}{T_P} = \left(\frac{1}{2}\right)\frac{dL}{L}$$

and $\Delta L = \alpha L_i \Delta T$, so, for temperature change dT,

$$dT_P = T_P\frac{1}{2}\frac{dL}{L} = T_P\frac{\alpha dT}{2}$$

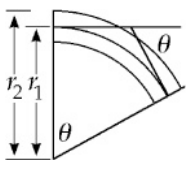
$$= 1.000\ \text{s}\ \frac{\left(19.0 \times 10^{-6}\,(°\text{C})^{-1}\right)(10.0°\text{C})}{2} = \boxed{9.50 \times 10^{-5}\ \text{s}}$$

(b) In one week, the time lost = 1 week (9.50×10^{-5} s lost per second)

$$\text{time lost} = (7.00\ \text{d/week})\left(\frac{86\,400\ \text{s}}{1.00\ \text{d}}\right)\left(9.50 \times 10^{-5}\ \frac{\text{s lost}}{\text{s}}\right)$$

$$\text{time lost} = \boxed{57.5\ \text{s lost}}$$

P19.60 The angle of bending θ, between tangents to the two ends of the strip, is equal to the angle the strip subtends at its center of curvature. (The angles are equal because their sides are perpendicular, right side to the right side and left side to left side.)



ANS. FIG. P19.60

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(a) The definition of radian measure gives $L_i + \Delta L_1 = \theta r_1$

and $L_i + \Delta L_2 = \theta r_2$. By subtraction,

$$\Delta L_2 - \Delta L_1 = \theta(r_2 - r_1)$$

$$\alpha_2 L_i \Delta T - \alpha_1 L_i \Delta T = \theta \Delta r$$

$$\boxed{\theta = \frac{(\alpha_2 - \alpha_1) L_i \Delta T}{\Delta r}}$$

(b) In the expression from part (a), θ is directly proportional to ΔT and also to $(\alpha_2 - \alpha_1)$. Therefore, θ is zero when either of these quantities becomes zero.

(c) The material that expands more when heated contracts more when cooled, so the bimetallic strip bends the other way. It is fun to demonstrate this with liquid nitrogen.

P19.61 (a) From ANS. FIG. P19.61, we see that the change in area is

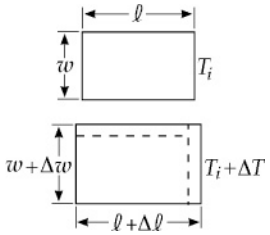
$$\Delta A = \ell \Delta w + w \Delta \ell + \Delta w \Delta \ell$$

Since $\Delta \ell$ and Δw are each small quantities, the product $\Delta w \Delta \ell$ will be very small. Therefore, we assume $\Delta w \Delta \ell \approx 0$.

Since $\Delta w = w \alpha \Delta T$ and $\Delta \ell = \ell \alpha \Delta T$,

we then have $\Delta A = \ell w \alpha \Delta T + \ell w \alpha \Delta T$,

and since $A = \ell w$, $\boxed{\Delta A = 2 \alpha A \Delta T}$



ANS. FIG. P19.61

(b) The approximation assumes $\Delta w \Delta \ell \approx 0$, or $\alpha \Delta T \approx 0$. Another way of stating this is $\boxed{\alpha \Delta T << 1}$.

P19.62 Let ρ_0 represent the density of the liquid at 0°C. At temperature T_C, the volume of a sample has changed according to $\Delta V = \beta V \Delta T = \beta V T_C$, so the density has become

$$\rho = \frac{m}{V + \beta V T_C} = \rho_0 \frac{1}{1 + \beta T_C}$$

so $$\rho(1 + \beta T_C) = \rho_0$$

Now the pressure at the bottom of the U tube is the same, whichever column it supports:

$$P_0 + \rho_0 g h_0 = P_0 + \rho g h_t$$

Simplifying,

$$\rho_0 h_0 = \rho h_t$$

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and substituting,

$$\rho(1 + \beta T_C)h_0 = \rho h_t$$

$$\left(1 + \beta T_C\right)h_0 = h_t \rightarrow \beta = \frac{1}{T_C}\left(\frac{h_t}{h_0} - 1\right)$$

P19.63 (a) Yes, so long as the coefficients of expansion remain constant.

(b) The coefficient of linear expansion of copper, $17.0 \times 10^{-6}\ °C^{-1}$, is greater than that of steel, $11.0 \times 10^{-6}\ °C^{-1}$, so the copper rod should start with a smaller length. Since the difference between the lengths of the two rods is to remain constant, we require

$$\Delta L_{Cu} = \Delta L_S$$

$$\alpha_{Cu} L_{Cu} \Delta T = \alpha_S L_S \Delta T$$

$$\left(17.0 \times 10^{-6}\ (°C)^{-1}\right) L_{Cu} \Delta T = \left(11.0 \times 10^{-6}\ (°C)^{-1}\right) L_S \Delta T$$

which gives

$$17.0 L_{Cu} = 11.0 L_S$$

Now, with $L_{Cu} + 5.00\ \text{cm} = L_S$ at 0°C, we obtain by substitution,

$$L_{Cu} + 5.00\ \text{cm} = \left(\frac{17.0}{11.0}\right) L_{Cu}$$

or $$L_{Cu} = \left(\frac{11.0}{6.00}\right)(5.00\ \text{cm}) = 9.17\ \text{cm}$$

With $L_S - L_C = 5.00$ cm, the only possibility is $L_S = 14.17$ cm and $L_C = 9.17$ cm.

P19.64 (a) Particle in equilibrium model

(b) On the piston,

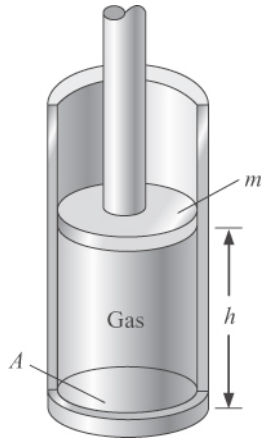
$$\sum F = F_{gas} - F_g - F_{air} = 0:$$

$$\sum F = PA - mg - P_0 A = 0$$

(c) In equilibrium, $P_{gas} = \frac{mg}{A} + P_0$.

Therefore, $\frac{nRT}{hA} = \frac{mg}{A} + P_0$,

or $$\boxed{h = \frac{nRT}{mg + P_0 A}},$$



ANS. FIG. P19.64

where we have used $V = hA$ as the volume of the gas.

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P19.65 We compute the moment of inertia from

$$I = \int r^2 dm$$

and since $r(T) = r(T_i)(1 + \alpha\Delta T)$,

$$\frac{I(T)}{I(T_i)} = (1 + \alpha\Delta T)^2$$

Thus $$\frac{\Delta I}{I} = \frac{I(T) - I(T_i)}{I(T_i)} = (1 + \alpha\Delta T)^2 - 1.$$

(a) With $\alpha = 17.0 \times 10^{-6}\,(°\text{C})^{-1}$ and $\Delta T = 100°\text{C}$, we find for Cu:

$$\frac{\Delta I}{I} = \left[1 + \left(17.0 \times 10^{-6}\,(°\text{C})^{-1}\right)(100°\text{C})\right]^2 - 1 = \boxed{0.340\%}$$

(b) With $\alpha = 24.0 \times 10^{-6}\,(°\text{C})^{-1}$ and $\Delta T = 100°\text{C}$, we find for Al:

$$\frac{\Delta I}{I} = \left[1 + \left(24.0 \times 10^{-6}\,(°\text{C})^{-1}\right)(100°\text{C})\right]^2 - 1 = \boxed{0.481\%}$$

P19.66 (a) Let m represent the sample mass. The number of moles is $n = \frac{m}{M}$ and the density is $\rho = \frac{m}{V}$. So $PV = nRT$ becomes $PV = \frac{m}{M}RT$ or $PM = \frac{m}{V}RT$.

Then, $\rho = \frac{m}{V} = \boxed{\frac{PM}{RT}}$

(b) $$\rho = \frac{PM}{RT} = \frac{\left(1.013 \times 10^5\ \text{N/m}^2\right)\left(0.032\,0\ \text{kg/mol}\right)}{\left(8.314\ \text{J/mol}\cdot\text{K}\right)(293\ \text{K})} = \boxed{1.33\ \text{kg/m}^3}$$

P19.67 After expansion, the length of one of the spans is

$$L_f = L_i(1 + \alpha\Delta T) = (125\ \text{m})\left[1 + 12 \times 10^{-6}\,(°\text{C})^{-1}(20.0°\text{C})\right]$$
$$= 125.03\ \text{m}$$

L_f, y, and the original 125-m length of this span form a right triangle with y as the altitude. Using the Pythagorean theorem gives

$$(125.03\ \text{m})^2 = y^2 + (125\ \text{m})^2$$

yielding $y = \boxed{2.74\ \text{m}}$.

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P19.68 Let $\ell = L/2$ represent the original length of one of the concrete slabs. After expansion, the length of each one of the spans is $\ell_f = \ell(1+\alpha\Delta T)$. Now, ℓ_f, y, and the original length ℓ of this span form a right triangle with y as the altitude. Using the Pythagorean theorem gives

$$\ell_f^2 = \ell^2 + y^2$$

or

$$y = \sqrt{\ell_f^2 - \ell^2} = \ell\sqrt{(1+\alpha\Delta T)^2 - 1} = (L/2)\sqrt{2\alpha\Delta T + (\alpha\Delta T)^2}$$

Since $\alpha\Delta T << 1$, we have $\boxed{y \approx L\sqrt{\alpha\Delta T/2}}$

P19.69 (a) Let V' represent the compressed volume at depth

$$B = \rho g V' \qquad P' = P_0 + \rho g d \qquad P'V' = P_0 V_i$$

$$B = \frac{\rho g P_0 V_i}{P'} = \boxed{\frac{\rho g P_0 V_i}{P_0 + \rho g d}}$$

(b) Since d is in the denominator, B must $\boxed{\text{decrease}}$ as the depth increases. (The volume of the balloon becomes smaller with increasing pressure.)

(c) To find the depth at which the buoyant force is half that at the surface, we write

$$\frac{1}{2} = \frac{B(d)}{B(0)} = \frac{\rho g P_0 V_i/(P_0 + \rho g d)}{\rho g P_0 V_i/P_0} = \frac{P_0}{P_0 + \rho g d}$$

Then, solve for d from $P_0 + \rho g d = 2P_0$:

$$d = \frac{P_0}{\rho g} = \frac{1.013\times10^5 \text{ N/m}^2}{(1.00\times10^3 \text{ kg/m}^3)(9.80 \text{ m/s}^2)} = \boxed{10.3 \text{ m}}$$

P19.70 (a) No torque acts on the disk so its angular momentum is constant. Yes: it increases. As the disk cools, its radius and, hence, its moment of inertia decrease. Conservation of angular momentum then requires that its angular speed increase.

(b) $$I_i\omega_i = I_f\omega_f = \frac{1}{2}MR_i^2\omega_i = \frac{1}{2}MR_f^2\omega_f = \frac{1}{2}M\left[R_i + R_i\alpha\Delta T\right]^2\omega_f$$

$$= \frac{1}{2}MR_i^2\left[1-\alpha|\Delta T|\right]^2\omega_f$$

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$$\omega_f = \omega_i\left[1-\alpha|\Delta T|\right]^{-2} = \frac{25.0\ \text{rad/s}}{\left[1-\left(17\times10^{-6}(°\text{C})^{-1}\right)(830°\text{C})\right]^2} = \frac{25.0\ \text{rad/s}}{0.972}$$

$$= \boxed{25.7\ \text{rad/s}}$$

P19.71 Visualize the molecules of various species all moving randomly. The net force on any section of wall is the sum of the forces of all of the molecules pounding on it.

For each gas alone, $P_1 = \frac{N_1 kT}{V}$ and $P_2 = \frac{N_2 kT}{V}$ and $P_3 = \frac{N_3 kT}{V}$, etc.

For all gases,

$$P_1V_1 + P_2V_2 + P_3V_3 \ldots \left(N_1 + N_2 + N_3 \ldots\right)kT \text{ and}$$
$$\left(N_1 + N_2 + N_3 \ldots\right)kT = PV$$

Also, $V_1 = V_2 = V_3 = \ldots = V$; therefore, $\boxed{P = P_1 + P_2 + P_3 \ldots}$

Challenge Problems

P19.72 (a) At 20.0°C, the unstretched lengths of the steel and copper wires are

$$L_s(20.0°\text{C}) = (2.000\ \text{m})\left[1+\left(11.0\times10^{-6}\ °\text{C}^{-1}\right)(-20.0°\text{C})\right]$$
$$= 1.999\,56\ \text{m}$$
$$L_c(20.0°\text{C}) = (2.000\ \text{m})\left[1+\left(17.0\times10^{-6}\ °\text{C}^{-1}\right)(-20.0°\text{C})\right]$$
$$= 1.999\,32\ \text{m}$$

Under a tension F, the length of the steel and copper wires are

$$L_s' = L_s\left[1+\frac{F}{YA}\right]_s \quad \text{and} \quad L_c' = L_c\left[1+\frac{F}{YA}\right]_c$$

where $L_s' + L_c' = 4.000\ \text{m}$

Since the tension F must be the same in each wire, we solve for F:

$$F = \frac{\left(L_s' + L_c'\right) - \left(L_s + L_c\right)}{L_s/Y_sA_s + L_c/Y_cA_c}$$

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When the wires are stretched, their areas become

$$A_s = \pi\left(1.000\times10^{-3}\text{ m}\right)^2\left[1+\left(11.0\times10^{-6}\ ^\circ\text{C}^{-1}\right)(-20.0)\right]^2$$
$$= 3.140\times10^{-6}\text{ m}^2$$
$$A_c = \pi\left(1.000\times10^{-3}\text{ m}\right)^2\left[1+\left(17.0\times10^{-6}\ ^\circ\text{C}^{-1}\right)(-20.0)\right]^2$$
$$= 3.139\times10^{-6}\text{ m}^2$$

Recall $Y_s = 20.0\times10^{10}$ Pa and $Y_c = 11.0\times10^{10}$ Pa. Substituting into the equation for F, we obtain

$$F = \left[4.000\text{ m} - \left(1.999\,56\text{ m} + 1.999\,32\text{ m}\right)\right] \times \frac{1}{\dfrac{1.999\,56\text{ m}}{\left(20.0\times10^{10}\text{ Pa}\right)\left(3.140\times10^{-6}\right)\text{ m}^2} + \dfrac{1.999\,32\text{ m}}{\left(11.0\times10^{10}\text{ Pa}\right)\left(3.139\times10^{-6}\right)\text{ m}^2}}$$

$$F = \boxed{125\text{ N}}$$

(b) To find the x coordinate of the junction,

$$L_s' = \left(1.999\,56\text{ m}\right)\left[1+\frac{125\text{ N}}{\left(20.0\times10^{10}\text{ N/m}^2\right)\left(3.140\times10^{-6}\text{ m}^2\right)}\right]$$
$$= 1.999958\text{ m}$$

Thus the x coordinate is $-2.000 + 1.999958 = \boxed{-4.20\times10^{-5}\text{ m}}$

P19.73 (a) We find the linear density from the volume density as the mass-per-volume multiplied by the volume-per-length, which is the cross-sectional area.

$$\mu = \frac{1}{4}\rho\left(\pi d^2\right) = \frac{1}{4}\pi(1.00\times10^{-3}\text{m})^2\left(7.86\times10^3\text{ kg/m}^3\right)$$
$$= \boxed{6.17\times10^{-3}\text{ kg/m}}$$

(b) Since $f_1 = \dfrac{v}{2L}$ and $v = \sqrt{\dfrac{T}{\mu}}$, then $f_1 = \dfrac{1}{2L}\sqrt{\dfrac{T}{\mu}}$, and we have for the tension

$$F = T = \mu\left(2Lf_1\right)^2 = \left(6.17\times10^{-3}\text{ kg/m}\right)\left[2(0.800\text{ m})\left(200\text{ s}^{-1}\right)\right]^2$$
$$= \boxed{632\text{ N}}$$

(c) At 0°C, the length of the guitar string will be

$$L_{\text{actual}} = L_{0^\circ\text{C}}\left(1+\frac{F}{AY}\right) = 0.800\text{ m}$$

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Where $L_{0°C}$ is the unstressed length at the low temperature. We know the string's cross-sectional area

$$A = \left(\frac{\pi}{4}\right)(1.00\times10^{-3}\ \text{m})^2 = 7.85\times10^{-7}\ \text{m}^2$$

and modulus $Y = 20.0 \times 10^{10}\ \text{N/m}^2$

Therefore,

$$\frac{F}{AY} = \frac{632\ \text{N}}{\left(7.85\times10^{-7}\ \text{m}^2\right)\left(20.0\times10^{10}\ \text{N/m}^2\right)} = 4.02\times10^{-3}$$

and $$L_{0°C} = \frac{0.800\ \text{m}}{1 + 4.02\times10^{-3}} = 0.796\ 8\ \text{m}$$

Then at 30°C, the unstressed length is

$$L_{30°C} = (0.796\ 8\ \text{m})\left[1+(30.0°\text{C})\left(11.0\times10^{-6}\ °\text{C}^{-1}\right)\right]$$
$$= 0.797\ 1\ \text{m}$$

With the same clamping arrangement,

$$0.800\ \text{m} = (0.797\ 1\ \text{m})\left[1+\frac{F'}{A'Y}\right]$$

where F' and A' are the new tension and the new (expanded) cross-sectional area. Then

$$\frac{F'}{A'Y} = \frac{0.800\ 0}{0.797\ 1} - 1 = 3.693\times10^{-3}$$

and

$$F' = A'Y(3.693\times10^{-3})$$

$$F' = (7.85\times10^{-7}\ \text{m}^2)(20.0\times10^{10}\ \text{N/m}^2)(3.693\times10^{-3})\,(1+\alpha\Delta T)^2$$

$$F' = (580\ \text{N})(1 + 3.30\times10^{-4})^2 = \boxed{580\ \text{N}}$$

(d) Also the new frequency f_1' is given by $\frac{f_1'}{f_1} = \sqrt{\frac{F'}{F}}$,

so $$f_1' = (200\ \text{Hz})\sqrt{\frac{580\ \text{N}}{632\ \text{N}}} = \boxed{192\ \text{Hz}}$$

P19.74 (a) $\frac{P_0V}{T} = \frac{P'V'}{T'}$ because the amount of gas remains constant.

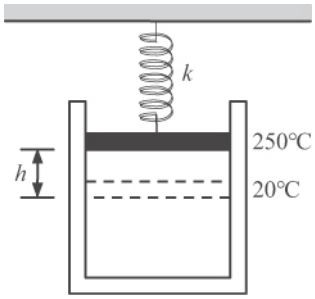
The volume increases by Ah when the piston rises:

$$V' = V + Ah$$

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When the piston compresses the piston by h, so the spring force increases by $F = kx = kh$, increasing the external pressure on the piston by kh/A:

$$V' = V + Ah$$



ANS. FIG. P19.74

Using the particle in equilibrium model applied to the piston,

$$\left(P_0 + \frac{kh}{A}\right)(V + Ah) = P_0 V\left(\frac{T'}{T}\right)$$

$$\left(1.013\times10^5\ \text{N/m}^2 + 2.00\times10^5\ \text{N/m}^3\, h\right)$$
$$\times\left(5.00\times10^{-3}\ \text{m}^3 + \left(0.010\ 0\ \text{m}^2\right)h\right)$$
$$= \left(1.013\times10^5\ \text{N/m}^2\right)\left(5.00\times10^{-3}\ \text{m}^3\right)\left(\frac{523\ \text{K}}{293\ \text{K}}\right)$$

$$2\ 000h^2 + 2\ 013h - 397 = 0$$

Taking the positive root, $h = \dfrac{-2\ 013 + 2\ 689}{4\ 000} = \boxed{0.169\ \text{m}}$

(b) $$P' = P + \frac{kh}{A} = 1.013\times10^5\ \text{Pa} + \frac{\left(2.00\times10^3\ \text{N/m}\right)(0.169\ \text{m})}{0.010\ 0\ \text{m}^2}$$

$$P' = \boxed{1.35\times10^5\ \text{Pa}}$$

P19.75 Each half of the spherical container is a particle in equilibrium. Therefore, using the result of Problem 14.58,

$$\sum F = 0 \ \rightarrow\ F_{\text{from gas}} = F_{\text{holding hemispheres together}}$$
$$\rightarrow\ P\left(\pi r^2\right) = \frac{F}{A}A = \sigma(2\pi rt) \ \rightarrow\ t = \frac{Pr}{2\sigma}$$

where σ is the yield strength of the steel. Find the mass of the steel sphere:

$$m_{\text{St}} = \rho_{\text{St}} V = \rho_{\text{St}}\left(4\pi r^2 t\right) = \rho_{\text{St}}\left[4\pi r^2\left(\frac{Pr}{2\sigma}\right)\right] = 2\pi\rho_{\text{St}}\frac{Pr^3}{\sigma}$$

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Find the pressure of the helium in the tank:

$$PV = nRT \quad \rightarrow \quad P = \frac{nRT}{V} = \frac{m_{\text{He}}}{M_{\text{He}}}\left(\frac{RT}{\frac{4}{3}\pi r^3}\right)$$

Substitute into the previous equation:

$$m_{\text{St}} = 2\pi\rho_{\text{St}}\frac{r^3}{\sigma}\left[\frac{m_{\text{He}}}{M_{\text{He}}}\left(\frac{RT}{\frac{4}{3}\pi r^3}\right)\right] = \frac{3}{2}\frac{\rho_{\text{St}}}{\sigma}\frac{m_{\text{He}}}{M_{\text{He}}}RT$$

Find the buoyant force on the balloon:

$$B = \rho_{\text{air}}gV_{\text{balloon}} = \rho_{\text{air}}g\frac{PV}{P_0} = \rho_{\text{air}}g\frac{nRT}{P_0} = \rho_{\text{air}}g\frac{m_{\text{He}}}{M_{\text{He}}}\frac{RT}{P_0}$$

where we use the pressure of helium in the balloon $P = P_0 =$ atmospheric pressure.

Find the net force on the balloon and tank:

$$\begin{aligned}\sum F &= B - m_{\text{He}}g - m_{\text{St}}g \\ &= \rho_{\text{air}}g\frac{m_{\text{He}}}{M_{\text{He}}}\frac{RT}{P_0} - m_{\text{He}}g - \frac{3}{2}\frac{\rho_{\text{St}}}{\sigma}\frac{m_{\text{He}}g}{M_{\text{He}}}RT \\ &= m_{\text{He}}g\left(\rho_{\text{air}}\frac{RT}{P_0M_{\text{He}}} - 1 - \frac{3}{2}\frac{\rho_{\text{St}}}{\sigma}\frac{RT}{M_{\text{He}}}\right) \\ &= m_{\text{He}}g\left[\frac{RT}{M_{\text{He}}}\left(\frac{\rho_{\text{air}}}{P_0} - \frac{3}{2}\frac{\rho_{\text{St}}}{\sigma}\right) - 1\right]\end{aligned}$$

Evaluate the brackets:

$$\begin{aligned}&\left[\frac{RT}{M_{\text{He}}}\left(\frac{\rho_{\text{air}}}{P_0} - \frac{3}{2}\frac{\rho_{\text{St}}}{\sigma}\right) - 1\right] \\ &\quad = \left\{\left[\frac{(8.314\ \text{J/mol}\cdot\text{K})(293\ \text{K})}{4 \times 10^{-3}\ \text{kg/mol}}\right.\right. \\ &\qquad \left.\left.\left(\frac{1.20\ \text{kg/m}^3}{1.013 \times 10^5\ \text{Pa}} - \frac{3}{2}\left(\frac{7\ 860\ \text{kg/m}^3}{5 \times 10^8\ \text{N/m}^2}\right)\right)\right] - 1\right\} \\ &\quad = \left\{\left[\left(6.09 \times 10^5\ \text{m}^2/\text{s}^2\right)\right.\right. \\ &\qquad \left.\left.\times\left(1.184\ 6 \times 10^{-5}\ \text{s}^2/\text{m}^2 - 2.358 \times 10^{-5}\ \text{s}^2/\text{m}^2\right)\right] - 1\right\} \\ &\quad = -8.146\end{aligned}$$

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Because the net force is negative, the balloon cannot lift the tank. If we can vary the strength of the steel, let's find out how strong the steel must be by evaluating σ to make the net force positive. We want the following to be true:

$$\frac{RT}{M_{\text{He}}}\left(\frac{\rho_{\text{air}}}{P_0} - \frac{3}{2}\frac{\rho_{\text{St}}}{\sigma}\right) - 1 > 0$$

Manipulating this inequality gives,

$$\frac{RT}{M_{\text{He}}}\left(\frac{\rho_{\text{air}}}{P_0} - \frac{3}{2}\frac{\rho_{\text{St}}}{\sigma}\right) > 1$$

$$\rightarrow \quad \frac{\rho_{\text{air}}}{P_0} - \frac{3}{2}\frac{\rho_{\text{St}}}{\sigma} > \frac{M_{\text{He}}}{RT} \quad \rightarrow \quad -\frac{3}{2}\frac{\rho_{\text{St}}}{\sigma} > \frac{M_{\text{He}}}{RT} - \frac{\rho_{\text{air}}}{P_0}$$

$$\rightarrow \quad \frac{3}{2}\frac{\rho_{\text{St}}}{\sigma} < -\frac{M_{\text{He}}}{RT} + \frac{\rho_{\text{air}}}{P_0} = \frac{-M_{\text{He}}P_0 + \rho_{\text{air}}RT}{P_0RT}$$

$$\rightarrow \quad \frac{2}{3}\frac{\sigma}{\rho_{\text{St}}} > \frac{P_0RT}{-M_{\text{He}}P_0 + \rho_{\text{air}}RT}$$

$$\rightarrow \quad \sigma > \frac{3}{2}\frac{\rho_{\text{St}}P_0RT}{-M_{\text{He}}P_0 + \rho_{\text{air}}RT}$$

$$= \frac{3}{2}\left[\frac{\left(7\ 860\ \text{kg/m}^3\right)\left(1.013 \times 10^5\ \text{Pa}\right)\left(8.314\ \text{J/mol}\cdot\text{K}\right)\left(293\ \text{K}\right)}{-\left(4 \times 10^{-3}\ \text{kg/mol}\right)\left(1.013 \times 10^5\ \text{Pa}\right) + \left(1.20\ \text{kg/m}^3\right)\left(8.314\ \text{J/mol}\cdot\text{K}\right)\left(293\ \text{K}\right)}\right]$$

$$\sigma = 11.6 \times 10^8\ \text{N/m}^2 = 2.3\sigma_{\text{actual}}$$

No, the steel would need to be 2.3 times stronger.

P19.76 With piston alone, T = constant, so $PV = P_0V_0$ or $P\left(Ah_i\right) = P_0\left(Ah_0\right)$.

With A = constant, $P = P_0\left(\frac{h_0}{h_i}\right)$.

But, $P = P_0 + \frac{m_pg}{A}$,

where m_p is the mass of the piston.

Thus $P_0 + \frac{m_pg}{A} = P_0\left(\frac{h_0}{h_i}\right)$, which reduces to $h_i = \frac{h_0}{1 + m_pg / P_0A}$.

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With the dog of mass M on the piston, a very similar calculation (replacing m_p by $m_p + M$) gives

$$h' = \frac{h_0}{1 + \left(m_p + M\right) g / P_0 A}$$

Thus, when the dog steps on the piston, it moves downward by

$$\Delta h = h_i - h'$$

$$= \frac{50.0 \text{ cm}}{1 + (20.0 \text{ kg})(9.80 \text{ m/s}^2) / \left[(1.013 \times 10^5 \text{ Pa}) \pi (0.400 \text{ m})^2\right]} - \frac{50.0 \text{ cm}}{1 + (45.0 \text{ kg})(9.80 \text{ m/s}^2) / \left[(1.013 \times 10^5 \text{ Pa}) \pi (0.400 \text{ m})^2\right]}$$

$$\Delta h = \boxed{2.38 \text{ mm}}$$

(b) $P = \text{const}$, so $\frac{V}{T} = \frac{V'}{T_i}$ or $\frac{Ah_i}{T} = \frac{Ah'}{T_i}$,

giving

$$T = T_i \left(\frac{h_i}{h'}\right) = T_i \frac{1 + \left(m_p + M\right) g / P_0 A}{1 + m_p g / P_0 A}$$

$$= 293 \text{ K} \, \frac{1 + (45.0 \text{ kg})(9.80 \text{ m/s}^2) / \left[(1.013 \times 10^5 \text{ Pa}) \pi (0.400 \text{ m})^2\right]}{1 + (20.0 \text{ kg})(9.80 \text{ m/s}^2) / \left[(1.013 \times 10^5 \text{ Pa}) \pi (0.400 \text{ m})^2\right]}$$

$$T = \boxed{294.4 \text{ K} = 21.4 \text{ °C}}$$

P19.77 (a) $\frac{dL}{L} = \alpha dT$: $\int_{T_i}^{T_i} \alpha \, dT = \int_{L_i}^{L_i} \frac{dL}{L} \Rightarrow \ln\left(\frac{L_f}{L_i}\right) = \alpha \Delta T \Rightarrow \boxed{L_f = L_i e^{\alpha \Delta T}}$

(b) $L_f = (1.00 \text{ m}) e^{\left[(2.00 \times 10^{-5} \text{ °C}^{-1})(100\text{°C})\right]} = 1.002\,002 \text{ m}$

$$L_f' = 1.00 \text{ m}\left[1 + \left(2.00 \times 10^{-5} \text{ °C}^{-1}\right)\left(100\text{°C}\right)\right] = 1.002\,000 :$$

$$\frac{L_f - L_f'}{L_f} = 2.00 \times 10^{-6} = \boxed{2.00 \times 10^{-4}\%}$$

(c) $L_f = (1.00 \text{ m}) e^{\left[(2.00 \times 10^{-2} \text{ °C}^{-1})(100\text{°C})\right]} = 7.389 \text{ m}$

$$L_f' = 1.00 \text{ m}\left[1 + \left(0.020\ 0 \text{°C}^{-1}\right)(100\text{°C})\right] = 3.000 \text{ m}$$

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$$\frac{L_f - L_f'}{L_f} = \boxed{59.4\%}$$

(d) P19.21 redone:

We start with

$$dV = \beta V dT \quad \rightarrow \quad \frac{dV}{V} = \beta dT \quad \rightarrow \quad V_f = V_i e^{\beta \Delta T}$$

where we assume β (and α) remains constant over the temperature range ΔT.

Thus, for the turpentine,

$$V_{\text{t, final}} = V_t e^{\beta_t \Delta T}$$

and for the aluminum cylinder, $V_{\text{Al, final}} = V_{\text{Al}} e^{3\alpha_{\text{Al}} \Delta T}$, where we assume α remains constant over the temperature range ΔT.

new (a):

$$\Delta V = V_t e^{\beta_t \Delta T} - V_{\text{Al}} e^{3\alpha_{\text{Al}} \Delta T}$$

$$\Delta V = V_i \left(e^{\beta_t \Delta T} - e^{3\alpha_{\text{Al}} \Delta T} \right)$$

$$= \left(2\ 000\ \text{cm}^3 \right) \left[e^{\left(9.00 \times 10^{-4}\ \text{C}^{-1}\right)(60.0°\text{C})} - e^{3\left(24.0 \times 10^{-6}°\text{C}^{-1}\right)(60.0°\text{C})} \right]$$

$$\Delta V = \boxed{102\ \text{mL of turpentine spills,}}$$

new (b):

The volume of the turpentine remaining in the cylinder at 80.0°C is the same as the volume of the aluminum cylinder at 80.0°C:

$$V_{t,\text{remaining}} = V_{\text{Al, final}} = V_{\text{Al}} e^{3\alpha_{\text{Al}} \Delta T}$$

$$= \left(2\ 000\ \text{cm}^3 \right) e^{3\left(24.0 \times 10^{-6}(°\text{C})^{-1}\right)(60.0°\text{C})}$$

$$= 2\ 009\ \text{cm}^3$$

$$\boxed{2.01\ \text{L remains in the cylinder at 80.0 °C}}$$

new (c):

The volume of turpentine at 80.0°C we found in part new (b), $V_{\text{t, remaining}} = V_{\text{Al}} e^{3\alpha_{\text{Al}} \Delta T}$, shrinks when the temperature changes by $-\Delta T$:

$$V_{\text{t, final}} = V_{\text{t, remaining}} e^{-\beta_t \Delta T} = V_{\text{Al}} e^{3\alpha_{\text{Al}} \Delta T} e^{-\beta_t \Delta T} = V_{\text{Al}} e^{(3\alpha_{\text{Al}} - \beta_t) \Delta T}$$

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$$V_{\text{t, final}} = V_{\text{Al}} e^{3\alpha_{\text{Al}}\Delta T} e^{-\beta_t \Delta T} = V_{\text{Al}} e^{(3\alpha_{\text{Al}} - \beta_t)\Delta T}$$

$$= \left(2\ 000\ \text{cm}^3\right) e^{\left[3\left(24.0\times10^{-6}\,°\text{C}^{-1}\right) - \left(9.00\times10^{-4}\,\text{C}^{-1}\right)\right](60.0°\text{C})}$$

$$V_{\text{t, final}} \simeq 1\ 903\ \text{cm}^3$$

Find the percentage of the cylinder that is empty at 20.0°C:

$$\frac{V_{\text{Al}} - V_{\text{Al}} e^{(3\alpha_{\text{Al}} - \beta_t)\Delta T}}{V_{\text{Al}} e} = 1 - e^{(3\alpha_{\text{Al}} - \beta_t)\Delta T}$$

Find the empty height of the cylinder above the turpentine:

$$\left(1 - e^{(3\alpha_{\text{Al}} - \beta_t)\Delta T}\right)(20.0\ \text{cm}) = 0.969\ \text{cm}$$

and the turpentine level at 20.0°C is 0.969 cm below the cylinder's rim.

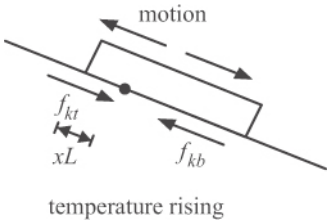
P19.78 (a) Let xL represent the distance of the stationary line below the top edge of the plate. The normal force on the lower part of the plate is $mg(1-x)\cos\theta$ and the force of kinetic friction on it is $\mu_k mg(1-x)\cos\theta$ up the roof. Again, $\mu_k mgx\cos\theta$ acts down the roof on the upper part of the plate. The near-equilibrium of the plate requires $\boxed{\sum F_x = 0}$.

$$-\mu_k mgx\cos\theta + \mu_k mg(1-x)\cos\theta - mg\sin\theta = 0$$

$$-2\mu_k mgx\cos\theta = mg\sin\theta - \mu_k mg\cos\theta$$

$$2\mu_k x = \mu_k - \tan\theta$$

$$x = \frac{1}{2} - \frac{\tan\theta}{2\mu_k}$$



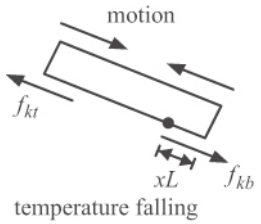
ANS. FIG. P19.78(a)

and the stationary line is indeed below the top edge by

$$xL = \frac{L}{2}\left(1 - \frac{\tan\theta}{\mu_k}\right)$$

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(b) With the temperature falling, the plate contracts faster than the roof. The upper part slides down and feels an upward frictional force $\mu_k mg(1-x)\cos\theta$. The lower part slides up and feels downward frictional force $\mu_k mgx\cos\theta$. The equation $\sum F_x = 0$ is then the same as in part (a) and the stationary line is above the bottom edge by $xL = \frac{L}{2}\left(1 - \frac{\tan\theta}{\mu_k}\right)$.



ANS. FIG. P19.78(b)

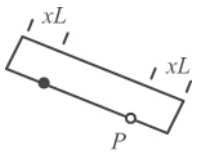
(c) Start thinking about the plate at dawn, as the temperature starts to rise. As in part (a), a line at distance xL below the top edge of the plate stays stationary relative to the roof as long as the temperature rises. The point P on the plate at distance xL above the bottom edge is destined to become the fixed point when the temperature starts falling. As the temperature rises, point P on the plate slides down the roof relative to the upper fixed line from $(L - xL - xL)$ to $(L - xL - xL)(1 + \alpha_2 \Delta T)$, a change of $\Delta L_{\text{plate}} = (L - xL - xL)\alpha_2 \Delta T$. The point on the roof originally under point P at the beginning of the expansion moves down not quite as much from $(L - xL - xL)$ to $(L - xL - xL)(1 + \alpha_1 \Delta T)$ relative to the upper fixed line; a change of $\Delta L_{\text{roof}} = L(1 - x - x)\alpha_1 \Delta T$. When the temperature drops, point P remains stationary on the roof while the roof contracts, pulling point P back by approximately ΔL_{roof}. Therefore, relative to the upper fixed line, point P has moved down the roof $\Delta L_{\text{plate}} - \Delta L_{\text{roof}}$. Its displacement for the day is

$$\Delta L = \Delta L_{\text{plate}} - \Delta L_{\text{roof}} = (\alpha_2 - \alpha_1)(L - xL - xL)\Delta T$$

$$= (\alpha_2 - \alpha_1)\left[L - 2\frac{L}{2}\left(1 - \frac{\tan\theta}{\mu_k}\right)\right](T_h - T_c)$$

$$= (\alpha_2 - \alpha_1)\left(\frac{L\tan\theta}{\mu_k}\right)(T_h - T_c)$$

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ANS. FIG. P19.78(c)

At dawn the next day the point P is farther down the roof by the distance ΔL. It represents the displacement of every other point on the plate.

(d) $$(\alpha_2 - \alpha_1)\left(\frac{L\tan\theta}{\mu_k}\right)(T_h - T_c)$$

$$= \left(24\times10^{-6}\ ^\circ\text{C}^{-1} - 15\times10^{-6}\ ^\circ\text{C}^{-1}\right) \times\left[\frac{(1.20\text{ m})\tan 18.5^\circ}{0.42}\right](32.0^\circ\text{C})$$

$$= \boxed{0.275\text{ mm}}$$

(e) If $\alpha_2 < \alpha_1$, the forces of friction reverse direction relative to parts (a) and (b) because the roof expands more than the plate as the temperature rises and less as the temperature falls. The diagram in part (a) then applies to temperature falling and the diagram in part (b) applies to temperature rising. A point on the plate xL from the top of the plate (which becomes the upper fixed line later when the plate contracts) moves upward from the lower fixed line by ΔL_{plate}, and when the temperature drops, the upper fixed line of the plate is carried down the roof by ΔL_{roof}, so the net change in the plate's position is $\Delta L_{\text{roof}} - \Delta L_{\text{plate}}$, same as before (up to a sign because now $\Delta L_{\text{roof}} > \Delta L_{\text{plate}}$).

The plate creeps down the roof each day by an amount given by the same expression (with α_2 and α_1 interchanged).

P19.79 See ANS. FIG. P19.79. Let 2θ represent the angle the curved rail subtends. We have

$$L_i + \Delta L = 2\theta R = L_i(1 + \alpha\Delta T)$$

and $$\sin\theta = \frac{L_i/2}{R} = \frac{L_i}{2R}.$$

Thus, $$\theta = \frac{L_i}{2R}(1+\alpha\Delta T) = (1+\alpha\Delta T)\sin\theta$$

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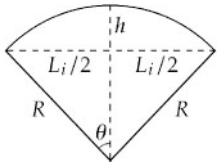
From Table 19.1, $\alpha = 11 \times 10^{-6} \left(^{\circ}\text{C}\right)^{-1}$, and ΔT = 25.0°C – 20.0°C = 5.00°C. We must solve the transcendental equation

$$\theta = (1 + \alpha \Delta T)\sin\theta = \left(1.000\ 005\ 5\right)\sin\theta$$

If your calculator is designed to solve such an equation, it may find the zero solution. Homing in on the nonzero solution gives, to five digits, $\theta = 0.018\ 165$ rad = 1.040 8°.

Now, $$h = R - R\cos\theta = \frac{L_i(1-\cos\theta)}{2\sin\theta}$$

This yields $\boxed{h = 4.54 \text{ m}}$, a remarkably large value compared to $\Delta L = 5.50$ cm.



ANS. FIG. P19.79

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ANSWERS TO EVEN-NUMBERED PROBLEMS

P19.2 (a) 31.7° C; (b) 31.7 K

P19.4 (a) −253° C; (b) −423° F

P19.6 (a) 56.7°C and –62.1°C; (b) 330 K and 211 K

P19.8 1.20 cm

P19.10 $\Delta r = 0.663$ mm to the right at 78.2° below the horizontal

P19.12 55.0°C

P19.14 1.58×10^{-3} cm

P19.16 0.548 gal

P19.18 Required $T = -376°$ C is below absolute zero.

P19.20 (a) 2.52×10^{6} N/m^2; (b) the concrete will not fracture

P19.22 (a) 396 N; (b) −101° C; (c) The original length divides out, so the answers would not change

P19.24 (a) $\dfrac{\rho_0}{1+\beta\Delta T}$; (b) m

P19.26 1.20 mol

P19.28 In each pump-up-and-discharge cycle, the volume of air in the tank doubles. Thus 1.00 L of water is driven out by the air injected at the first pumping, 2.00 L by the second, and only the remaining 1.00 L by the third. Each person could more efficiently use his device by starting with the tank half full of water, instead of 80% full.

P19.30 (a) 1.17×10^{-3} kg; (b) 11.5 mN; (c) 1.01 kN; (d) molecules must be moving very fast

P19.32 4.39 kg

P19.34 6.64×10^{-27} kg

P19.36 2.42×10^{11} molecules

P19.38 7.13 m

P19.40 $m_1 - m_2 = \dfrac{P_0 VM}{R}\left(\dfrac{1}{T_1} - \dfrac{1}{T_2}\right)$

P19.42 ~10^2 kg

P19.44 $m_f = m_i\left(\dfrac{P_{gf} + P_0}{P_{gi} + P_0}\right)$

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P19.46 35.016 m

P19.48 2.86 atm

P19.50 95.0°; T falls below 100°C, so steam condenses, and the expensive apparatus falls (assuming that the boiling point does not change significantly with the change in pressure).

P19.52 $\Delta h = (\beta - 3\alpha)\dfrac{V\Delta T}{A}$

P19.54 (a) $\theta = 2\sin^{-1}\left(\dfrac{1+\alpha_{\text{Al}}T_{\text{C}}}{2}\right)$; (b) Yes; (c) Yes,

(d) $\theta = 2\sin^{-1}\left(\dfrac{1+\alpha_{\text{Al}}T_{\text{C}}}{2\left(1+\alpha_{\text{invar}}T_{\text{C}}\right)}\right)$; (e) 61.0°; (f) 59.6°

P19.56 0.523 kg

P19.58 (a) $\beta = \dfrac{1}{T}$; (b) $3.66 \times 10^{-3}\ \text{K}^{-1}$; (c) $\beta_{\text{He}} = 3.665 \times 10^{-3}\ \text{K}^{-1}$, this agrees within 0.06% of the tabulated value; (d) $\beta_{\text{He}} = 3.67 \times 10^{-3}\ \text{K}^{-1}$, this agrees within 0.2% of the tabulated value

P19.60 $\theta = \dfrac{(\alpha_2 - \alpha_1)L_i\Delta T}{\Delta r}$; (b) In the expression from part (a), θ is directly proportional to ΔT and also to $(\alpha_2 - \alpha_1)$. Therefore, θ is zero when either of these quantities becomes zero; (c) the bimetallic strip bends the other way

P19.62 See P19.62 for the full solution.

P19.64 (a) Particle in equilibrium model; (b) On the piston,

$\sum F = F_{\text{gas}} - F_{\text{g}} - F_{\text{air}} = 0 : \sum F = PA - mg - P_0A = 0$; (c) $h = \dfrac{nRT}{mg + P_0A}$

P19.66 (a) $\dfrac{PM}{RT}$; (b) $1.33\ \text{kg/m}^3$

P19.68 $y \approx L\sqrt{\alpha\Delta T/2}$

P19.70 (a) No torque acts on the disk so its angular momentum is constant. Yes: it increases. As the disk cools, its radius, and hence, its moment of inertia decrease. Conservation of angular momentum then requires that its angular speed increase; (b) 25.7 rad/s

P19.72 (a) 125 N; (b) -4.20×10^{-5} m

P19.74 (a) 0.169 m; (b) 1.35×10^5 Pa

P19.76 (a) 2.38 mm; (b) 294.4 K = 21.4° C

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P19.78 (a) $\sum F_x = 0$; (b) With the temperature falling, the plate contracts faster than the roof. The upper part slides down and feels an upward frictional force $\mu_k mg(1-x)\cos\theta$. The lower part slides up and feels downward frictional force $\mu_k mgx\cos\theta$. The equation $\sum F_x = 0$ is then the same as in part (a), and the stationary line is above the bottom edge by $xL = \frac{L}{2}\left(1 - \frac{\tan\theta}{\mu_k}\right)$; (c) See P19.78(c) for the full explanation; (d) 0.275 mm; (e) The plate creeps down the roof each day by an amount given by the same expression (with α_2 and α_1 interchanged).

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